Báo cáo vòng 8: hiệu chuẩn nhân quả, độ bền bộ lọc, off-map tự nhiên và HMM liên hợp

Branch `hmm-navigation-5`, 2026-11-01. Nhật ký `RESEARCH_LOG.md` Entry 30–31 (đăng ký trước, push trước đánh giá). Lý thuyết: `paper/cms_theory.md` §4c–4d. Vòng trước: `paper/BAO_CAO_E29.pdf`.

Tóm tắt.

1. **Phương pháp đề xuất giữ nguyên và được củng cố: CMS-CPC/H+SH+G+NGRr** (= luật vòng 7, cổng điều hướng mở khoá **giới hạn tốc độ**). LODO trên mọi luật không nhãn của vòng này (gồm cả hai biến thể cổng và mọi nhóm hiệu chuẩn) chọn nó ở **7/7 fold** (α = 0.05; ở α = 0.01 bản α = 0.01 của nó 6/7).
2. **Bản triển khai được (nhân quả, không nhìn tương lai của phiên): CMS-CPCW/H+SH[both]+NGRr** hơn luật tốt nhất trước CMS (CPC-SC-T½+SH) cả **recall 7/7 và sai số điều hướng 7/7** trên 366 run (CI bootstrap cụm: recall 6/7, sai số 7/7), segFAR trung bình ≤ 0.008; bản nhân quả hợp lệ ở 7/7 dataset cả α = 0.05 lẫn 0.01 (Định lý 6″, Mệnh đề 12).
3. **Độ bền bộ lọc:** dưới cả ba bộ lọc thay thế (nhiễu quá trình ×4; thêm σ fix 10 m; khởi tạo không nhãn), phương pháp đề xuất hơn CPC-SC-T½+SH về recall 7/7 và sai số 7/7 ở cả ba (CI: recall 6/7 mỗi cấu hình; sai số 5/7, 7/7, 6/7).
4. **Off-map tự nhiên** (4Seasons chéo cảnh, 36 run, phiên hoàn toàn off-map): mọi biến thể CMS hợp lệ ở α = 0.05 (không run nào vượt α), recall cảnh A cao hơn luật cũ **6/6 cặp** (0.71 so với 0.59 gộp) và sai số thấp hơn (18.8 so với 26.0 m); baseline có nhãn vượt α ở 11–28 % run. Bản chuyển đạo vượt α = 0.01 ở 2/36 run — đúng chế độ hỏng π = 0 lý thuyết dự báo, bản nhân quả thì không.
5. **Hai phát hiện về cổng** (báo cáo thẳng): code vòng 7 khác mô tả (mở khoá giới hạn tốc độ thay vì mở cả chuỗi); bản sửa "mở cả chuỗi" đăng ký trước hoá ra **kém hơn** (Boreas +2.0 đến +2.9 m, CI > 0) ⇒ giữ cả hai, LODO chọn bản giới hạn tốc độ.
6. **HMM là trung tâm (Entry 31):** HMM liên hợp vị trí–hiện diện với trạng thái OFF có emission hiệu chuẩn phản thực; biên OFF của nó **trùng chính xác** chuỗi hiện diện đã hiệu chuẩn (Mệnh đề 13), nên mọi định lý là phát biểu về chính HMM. Trên 3 run thiết kế, ghép nối liên hợp giữ nguyên quyết định (cùng biên OFF) và chữa lock-in của cùng chuỗi vị trí (Nordland: sai số CMS-CPCW 27.3 → 6.0 m, độ đúng vị trí sau lỗ 0.80 → 0.90) nhưng **không tăng recall** so với chuỗi vị trí gốc (−0.01, do emission dựng lại từ top-K); đánh giá 114 run đang chạy (§5).

1. Câu hỏi của vòng này (không có robot thật, chỉ benchmark trên dataset)

#	Vấn đề còn lại sau vòng 7	Ảnh hưởng	Thí nghiệm
1	Hiệu chuẩn CMS theo phiên là **chuyển đạo** (dùng cả frame tương lai của phiên)	Không triển khai được nguyên dạng; reviewer sẽ hỏi	Nhóm `[online]` (phiên khác + phần đã qua của phiên), `[both]` (giao phiên ∧ phiên khác)
2	Cổng điều hướng và sai số gắn với **một** cấu hình Kalman khởi tạo ở pose thật	Kết luận về sai số có thể phụ thuộc bộ lọc	F1 nhiễu quá trình ×4, F2 = F1 + σ fix 10 m, F3 khởi tạo không nhãn (top-1 frame đầu)
3	Off-map hầu hết là **lỗ tổng hợp** (trừ NCLT); phiên **hoàn toàn off-map** có π = 0	Bảo đảm của nhóm `cond` vô nghĩa ở phiên đó	Benchmark 4Seasons chéo cảnh: bản đồ cảnh A, truy vấn cảnh B (mọi frame off-map, ảnh và VIO thật)
4	Chốt luật chính	—	CMS-CPCW/H+SH+NGR (định lý chặt) và CMS-CPC/H+SH+G+NGR (recall), khoảng tin cậy ghép cặp, LODO

2. Phương pháp và sửa lỗi của vòng này

Hiệu chuẩn nhân quả (`analysis/cms.py: online_parts`). `[online]`: tại frame t, ngưỡng khởi phát là phân vị (1 − α) của max log-odds trên các đoạn phản thực của **các phiên khác** cộng các đoạn H frame **đã kết thúc trước t** của phiên hiện tại; phép biến đổi p → e và a của Beta-EM cố định từ các phiên khác. Định lý 6″: tỷ lệ khởi phát sai mỗi đoạn ≤ α(n+1)/(πn+1) với π của tập hiệu chuẩn có trước đoạn đó — một phiên hoàn toàn off-map vẫn được bảo đảm nhờ phần on-map của các phiên khác. `[both]`: báo khi luật theo phiên **và** luật theo phiên khác cùng báo; Mệnh đề 12: segFAR ≤ min của hai luật (hợp lệ nếu một trong hai null hợp lệ).

Cổng +NGR: một chỗ code khác mô tả, một sửa, hai biến thể (tìm ra khi thiết kế F3, Nordland CosPlace s0). (a) Mở khoá kiểm lại độ đồng nhất của innovation sau mỗi fix được nhận — chính bước cập nhật của bộ lọc làm innovation nhảy ⇒ chuỗi đã mở chỉ được nhận **một fix mỗi R = 10 frame** (khác mô tả "mở cho cả chuỗi" của vòng 7). Đăng ký trước: sửa thành mở đến hết chuỗi (`+NGR`). Giữa đánh giá (202/366 run, ghi nhật ký trước mọi phân tích khác) thấy sửa này **lẫn lộn** với khởi tạo pose thật: Boreas +1.3 đến +2.6 m (một chuỗi báo sai đồng nhất lọt hết khi đã mở), CMU +0.5 m, còn Nordland / 4S thấp hơn. Hành vi vòng 7 hoá ra là một thiết kế hợp lệ — **mở giới hạn tốc độ**, chặn tác hại của chuỗi báo sai đồng nhất — nên giữ cả hai (`+NGRr` = vòng 7 + sàn bất định), tính lại mọi run cho `+NGRr`, và để LODO chọn. (b) Bán kính cổng max(τ, 3√P⁻): khi bộ lọc chưa định vị (P⁻ lớn), fix đúng có innovation cỡ √P⁻; với khởi tạo pose thật sàn này gần như không hoạt động (`+NGRr` trùng tuyệt đối với +NGR vòng 7 trên 360/366 run; 6 run NCLT lệch ≤ 0.001 recall).

Benchmark off-map tự nhiên (`analysis/natural_eval.py`). 6 cặp có thứ tự của {office loop, old town, countryside} × 6 descriptor = 36 run. Bản đồ = bản ghi tham chiếu của A (không lỗ); truy vấn = các bản ghi còn lại của A rồi mọi bản ghi của B. Recall tính trên A, segFAR trên các đoạn off-map (gần như toàn bộ là B), sai số điều hướng chỉ tính trên A (toạ độ của B không liên quan bản đồ). Baseline có nhãn được huấn luyện trên bản ghi A dễ nhất **và một bản ghi B có nhãn** (có lợi cho baseline).

3. Tối ưu tính toán

- Ba đợt Kaggle, lấp đầy từng acc trước khi sang acc khác (2 phiên L4×4 + 5 phiên CPU mỗi acc, 10 acc):
- Đợt 1: 20 phiên L4 (đánh giá chính 366 run + F3) và 50 phiên CPU (F1, F2: 732 job).
- Đợt 2: 48 phiên CPU (36 job benchmark tự nhiên, mỗi job một pipeline đầy đủ ~21 phút; 72 job bản đồ khuyết).
- Đợt 3: 20 phiên L4 + 50 phiên CPU (biến thể cổng giới hạn tốc độ cho mọi cấu hình).
- Chính sách L4 của user (kẹt > 5 phút thì huỷ + đẩy lại, tối đa 5 lần, sau đó 2×T4): 7/20 phiên được đẩy lại 1–3 lần, không phiên nào phải xuống T4.
- Tái dùng: thống kê phản thực chính xác của 216 run bản đồ khuyết và cache của benchmark tự nhiên được đóng gói thành dataset Kaggle (`hmm-holes-cms-exact`, `hmm-natural-cache`), các lần chạy lại chỉ còn phần quyết định (CPU, vài phút).
- Kiểm tái lập: các luật không cổng của vòng này trùng **tuyệt đối** với vòng 7 trên mọi run (366/366); `+NGRr` trùng tuyệt đối với +NGR của vòng 7 trên 360/366 run — 6 run NCLT (bộ lọc EKF có trạng thái hướng, √P lớn hơn) lệch ≤ 0.001 recall / ≤ 0.09 m do sàn bất định 3√P⁻.

4. Kết quả

Bảng 1. Hiệu chuẩn theo nhóm, α = 0.05: recall / segFAR TB / segFAR run tệ nhất / sai số điều hướng (m), 366 run

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
Xu'21 (nhãn)	0.684 / 0.043 / 0.567 / 25.3	0.526 / 0.021 / 0.065 / 13.7	0.686 / 0.227 / 1.000 / 18.1	0.740 / 0.038 / 0.113 / 59.3	0.724 / 0.215 / 0.997 / 21.2	0.625 / 0.101 / 1.000 / 70.9	0.254 / 0.088 / 0.983 / 81.0
Claxton'24 (nhãn)	0.741 / 0.116 / 0.619 / 37.3	0.663 / 0.213 / 0.425 / 33.2	0.874 / 0.167 / 0.818 / 14.4	0.521 / 0.267 / 0.442 / 88.8	0.798 / 0.154 / 0.589 / 18.9	0.724 / 0.166 / 0.544 / 90.3	0.597 / 0.213 / 0.498 / 42.9
TAO'26 (nhãn)	0.801 / 0.240 / 0.830 / 46.2	0.756 / 0.707 / 1.000 / 57.6	0.930 / 0.370 / 1.000 / 38.4	0.805 / 0.579 / 1.000 / 187.9	0.935 / 0.740 / 1.000 / 54.0	0.839 / 0.557 / 1.000 / 168.4	0.769 / 0.732 / 1.000 / 95.1

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CPC-SC+CS (vòng 3)	0.661 / 0.001 / 0.021 / 5.1	0.537 / 0.000 / 0.000 / 4.6	0.731 / 0.002 / 0.019 / 3.8	0.644 / 0.000 / 0.003 / 9.6	0.709 / 0.000 / 0.000 / 11.6	0.600 / 0.000 / 0.010 / 30.8	0.297 / 0.000 / 0.007 / 53.1
CPC-SC-T½+SH (vòng 5)	0.666 / 0.002 / 0.023 / 5.7	0.588 / 0.001 / 0.003 / 4.9	0.789 / 0.018 / 0.188 / 5.2	0.713 / 0.001 / 0.010 / 9.5	0.738 / 0.000 / 0.003 / 11.4	0.640 / 0.007 / 0.034 / 29.8	0.334 / 0.000 / 0.015 / 51.3
CPC-SC-T½+SH+NGRr	0.666 / 0.001 / 0.017 / 5.3	0.588 / 0.000 / 0.000 / 3.9	0.789 / 0.011 / 0.111 / 4.0	0.712 / 0.000 / 0.003 / 9.5	0.738 / 0.000 / 0.003 / 11.5	0.629 / 0.006 / 0.034 / 31.2	0.325 / 0.000 / 0.004 / 52.2
CPC-SC-T½+SH+NGR	0.666 / 0.001 / 0.017 / 5.4	0.588 / 0.000 / 0.000 / 3.9	0.789 / 0.011 / 0.125 / 4.4	0.713 / 0.000 / 0.003 / 9.4	0.738 / 0.000 / 0.003 / 11.5	0.633 / 0.006 / 0.034 / 30.5	0.328 / 0.000 / 0.004 / 52.3
CMS-CPCW/ H+SH [phiên (chuyển đạo)]	0.766 / 0.004 / 0.028 / 10.0	0.611 / 0.005 / 0.029 / 5.1	0.827 / 0.013 / 0.065 / 6.4	0.773 / 0.002 / 0.018 / 8.3	0.799 / 0.001 / 0.015 / 9.8	0.710 / 0.008 / 0.051 / 27.1	0.479 / 0.001 / 0.010 / 35.6
CMS-CPCW/ H+SH+NGRr [phiên (chuyển đạo)]	0.766 / 0.002 / 0.022 / 5.5	0.610 / 0.001 / 0.007 / 4.8	0.826 / 0.009 / 0.054 / 4.1	0.772 / 0.000 / 0.010 / 5.8	0.798 / 0.000 / 0.006 / 9.7	0.700 / 0.007 / 0.040 / 26.3	0.468 / 0.000 / 0.007 / 37.3
CMS-CPCW/ H+SH+NGR [phiên (chuyển đạo)]	0.766 / 0.002 / 0.022 / 8.4	0.610 / 0.001 / 0.007 / 4.6	0.826 / 0.009 / 0.054 / 4.7	0.772 / 0.000 / 0.010 / 5.9	0.798 / 0.000 / 0.006 / 9.7	0.704 / 0.007 / 0.040 / 26.7	0.472 / 0.000 / 0.007 / 36.8
CMS-CPCW/ H+SH [phiên khác]	0.745 / 0.004 / 0.028 / 8.3	0.604 / 0.004 / 0.016 / 5.1	0.823 / 0.012 / 0.059 / 6.3	0.737 / 0.004 / 0.041 / 12.1	0.770 / 0.001 / 0.018 / 10.1	0.702 / 0.009 / 0.062 / 26.6	0.475 / 0.003 / 0.026 / 40.1
CMS-CPCW/ H+SH+NGRr [phiên khác]	0.745 / 0.002 / 0.022 / 5.1	0.603 / 0.000 / 0.000 / 3.7	0.820 / 0.008 / 0.050 / 4.0	0.735 / 0.001 / 0.011 / 7.6	0.769 / 0.000 / 0.012 / 10.1	0.692 / 0.007 / 0.051 / 27.3	0.465 / 0.002 / 0.022 / 38.8
CMS-CPCW/ H+SH+NGR [phiên khác]	0.745 / 0.002 / 0.022 / 5.5	0.603 / 0.000 / 0.000 / 3.7	0.820 / 0.008 / 0.050 / 4.6	0.736 / 0.001 / 0.013 / 7.8	0.769 / 0.000 / 0.012 / 10.1	0.695 / 0.007 / 0.051 / 27.1	0.469 / 0.002 / 0.022 / 40.2
CMS-CPCW/ H+SH [online nhân quả]	0.728 / 0.004 / 0.028 / 6.9	0.605 / 0.002 / 0.010 / 4.9	0.822 / 0.012 / 0.060 / 6.2	0.696 / 0.002 / 0.013 / 12.2	0.754 / 0.001 / 0.021 / 10.3	0.691 / 0.009 / 0.062 / 26.9	0.467 / 0.002 / 0.018 / 40.9
CMS-CPCW/ H+SH+NGRr [online nhân quả]	0.728 / 0.002 / 0.021 / 5.0	0.604 / 0.000 / 0.000 / 3.7	0.820 / 0.008 / 0.050 / 4.0	0.695 / 0.000 / 0.008 / 8.8	0.753 / 0.000 / 0.012 / 10.3	0.681 / 0.007 / 0.051 / 28.0	0.458 / 0.001 / 0.014 / 40.3
CMS-CPCW/ H+SH+NGR [online nhân quả]	0.728 / 0.002 / 0.021 / 5.4	0.604 / 0.000 / 0.000 / 3.7	0.820 / 0.009 / 0.050 / 4.5	0.695 / 0.000 / 0.008 / 8.6	0.753 / 0.000 / 0.012 / 10.3	0.684 / 0.007 / 0.051 / 27.7	0.461 / 0.001 / 0.018 / 41.1
CMS-CPCW/ H+SH [giao phiên ∧ phiên khác]	0.733 / 0.003 / 0.027 / 7.2	0.600 / 0.001 / 0.007 / 4.6	0.815 / 0.010 / 0.056 / 5.6	0.723 / 0.001 / 0.010 / 8.1	0.767 / 0.000 / 0.006 / 10.1	0.692 / 0.007 / 0.051 / 27.0	0.427 / 0.001 / 0.010 / 43.0

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPCW/ H+SH+NGRr [giao phiên ∧ phiên khác]	0.733 / 0.002 / 0.021 / 4.9	0.599 / 0.000 / 0.000 / 3.7	0.815 / 0.008 / 0.050 / 3.9	0.721 / 0.000 / 0.003 / 7.3	0.766 / 0.000 / 0.003 / 10.1	0.682 / 0.005 / 0.040 / 27.8	0.417 / 0.000 / 0.007 / 42.4
CMS-CPCW/ H+SH+NGR [giao phiên ∧ phiên khác]	0.733 / 0.002 / 0.021 / 5.2	0.599 / 0.000 / 0.000 / 3.7	0.815 / 0.008 / 0.050 / 4.3	0.722 / 0.000 / 0.003 / 7.1	0.766 / 0.000 / 0.003 / 10.1	0.685 / 0.005 / 0.040 / 27.5	0.421 / 0.000 / 0.007 / 43.1
CMS-CPC/ H+SH+G [phiên (chuyển đạo)]	0.776 / 0.005 / 0.030 / 11.3	0.614 / 0.008 / 0.042 / 6.8	0.823 / 0.017 / 0.067 / 6.5	0.767 / 0.004 / 0.020 / 7.3	0.801 / 0.002 / 0.018 / 9.9	0.713 / 0.008 / 0.062 / 26.8	0.503 / 0.002 / 0.026 / 33.7
CMS-CPC/ H+SH+G+NGR [phiên (chuyển đạo)]	0.775 / 0.003 / 0.022 / 4.5	0.612 / 0.002 / 0.007 / 4.3	0.822 / 0.009 / 0.050 / 4.0	0.765 / 0.000 / 0.000 / 5.6	0.800 / 0.000 / 0.006 / 9.7	0.703 / 0.006 / 0.051 / 26.1	0.491 / 0.001 / 0.010 / 34.3
CMS-CPC/ H+SH+G+NGR0 [phiên (chuyển đạo)]	0.775 / 0.003 / 0.022 / 6.6	0.613 / 0.002 / 0.010 / 4.9	0.822 / 0.010 / 0.050 / 4.5	0.766 / 0.000 / 0.000 / 5.5	0.800 / 0.000 / 0.006 / 9.7	0.706 / 0.006 / 0.051 / 25.8	0.494 / 0.001 / 0.010 / 33.3
CMS-CPC/ H+SH+G [phiên khác]	0.764 / 0.006 / 0.029 / 7.5	0.610 / 0.008 / 0.042 / 6.6	0.816 / 0.016 / 0.065 / 6.3	0.751 / 0.006 / 0.037 / 9.7	0.779 / 0.001 / 0.024 / 10.0	0.710 / 0.009 / 0.062 / 27.8	0.494 / 0.003 / 0.026 / 36.4
CMS-CPC/ H+SH+G+NGR0 [phiên khác]	0.764 / 0.003 / 0.022 / 4.3	0.608 / 0.002 / 0.010 / 4.3	0.816 / 0.009 / 0.054 / 4.0	0.749 / 0.000 / 0.003 / 5.6	0.777 / 0.000 / 0.009 / 9.9	0.699 / 0.006 / 0.051 / 26.9	0.483 / 0.001 / 0.013 / 35.7
CMS-CPC/ H+SH+G+NGR [phiên khác]	0.764 / 0.003 / 0.022 / 4.7	0.609 / 0.002 / 0.013 / 4.9	0.816 / 0.009 / 0.054 / 4.5	0.749 / 0.000 / 0.006 / 5.9	0.777 / 0.000 / 0.009 / 9.9	0.702 / 0.006 / 0.051 / 27.0	0.486 / 0.001 / 0.013 / 35.9
CMS-CPC/ H+SH+G [online nhân quả]	0.740 / 0.004 / 0.028 / 5.9	0.599 / 0.005 / 0.033 / 5.9	0.812 / 0.014 / 0.065 / 6.1	0.684 / 0.002 / 0.015 / 11.8	0.757 / 0.000 / 0.018 / 10.3	0.690 / 0.007 / 0.062 / 27.1	0.476 / 0.002 / 0.023 / 38.7
CMS-CPC/ H+SH+G+NGR0 [online nhân quả]	0.740 / 0.002 / 0.021 / 4.5	0.598 / 0.001 / 0.003 / 4.2	0.812 / 0.009 / 0.054 / 3.9	0.682 / 0.000 / 0.003 / 8.8	0.756 / 0.000 / 0.009 / 10.2	0.679 / 0.005 / 0.051 / 28.1	0.465 / 0.001 / 0.007 / 37.6
CMS-CPC/ H+SH+G+NGR [online nhân quả]	0.740 / 0.002 / 0.021 / 4.8	0.598 / 0.001 / 0.007 / 4.8	0.812 / 0.009 / 0.054 / 4.4	0.683 / 0.000 / 0.006 / 8.9	0.756 / 0.000 / 0.009 / 10.2	0.682 / 0.005 / 0.051 / 27.9	0.469 / 0.001 / 0.007 / 38.1
CMS-CPC/ H+SH+G [giao phiên ∧ phiên khác]	0.751 / 0.005 / 0.028 / 6.9	0.605 / 0.005 / 0.033 / 6.9	0.812 / 0.014 / 0.060 / 5.7	0.736 / 0.002 / 0.011 / 7.6	0.776 / 0.001 / 0.012 / 10.0	0.702 / 0.007 / 0.062 / 26.5	0.461 / 0.001 / 0.026 / 39.6
CMS-CPC/ H+SH+G+NGR0 [giao phiên ∧ phiên khác]	0.751 / 0.002 / 0.021 / 4.3	0.604 / 0.001 / 0.007 / 4.3	0.812 / 0.009 / 0.052 / 3.8	0.734 / 0.000 / 0.000 / 6.3	0.775 / 0.000 / 0.003 / 10.0	0.691 / 0.005 / 0.051 / 26.8	0.450 / 0.000 / 0.010 / 38.2

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPC/ H+SH+G+NGR [giao phiên ∧ phiên khác]	0.751 / 0.002 / 0.021 / 4.7	0.604 / 0.002 / 0.010 / 4.9	0.812 / 0.009 / 0.052 / 4.2	0.734 / 0.000 / 0.000 / 6.1	0.775 / 0.000 / 0.003 / 10.0	0.695 / 0.005 / 0.051 / 26.5	0.454 / 0.000 / 0.010 / 38.4
Oracle	0.88 / – / – / 2.1	0.72 / – / – / 2.2	0.94 / – / – / 3.0	0.67 / – / – / 4.8	0.91 / – / – / 8.3	0.84 / – / – / 19.4	0.73 / – / – / 16.5

Bảng 2. Như Bảng 1, α = 0.01

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPC/ H+SH+NGR	0.780 / 0.003 / 0.022 / 9.1	0.628 / 0.004 / 0.023 / 4.9	0.833 / 0.010 / 0.058 / 4.9	0.797 / 0.002 / 0.015 / 6.9	0.804 / 0.000 / 0.006 / 9.6	0.712 / 0.008 / 0.051 / 26.6	0.516 / 0.001 / 0.016 / 33.6
CMS-CPC/ H+SH+NGR, α = 0.01	0.751 / 0.001 / 0.014 / 5.3	0.594 / 0.001 / 0.003 / 4.1	0.794 / 0.007 / 0.042 / 3.7	0.722 / 0.001 / 0.015 / 8.5	0.730 / 0.000 / 0.003 / 10.6	0.694 / 0.007 / 0.040 / 27.5	0.403 / 0.001 / 0.007 / 43.8
CMS-CPCW/ H+SH [phiên (chuyển đạo)]	0.733 / 0.001 / 0.016 / 4.7	0.551 / 0.000 / 0.000 / 5.5	0.781 / 0.006 / 0.042 / 4.1	0.589 / 0.000 / 0.005 / 9.7	0.677 / 0.000 / 0.003 / 11.8	0.674 / 0.006 / 0.045 / 27.9	0.349 / 0.001 / 0.010 / 56.2
CMS-CPCW/ H+SH+NGRr [phiên (chuyển đạo)]	0.733 / 0.001 / 0.011 / 4.4	0.551 / 0.000 / 0.000 / 4.6	0.781 / 0.006 / 0.042 / 3.6	0.586 / 0.000 / 0.000 / 10.2	0.676 / 0.000 / 0.003 / 11.8	0.664 / 0.006 / 0.034 / 28.8	0.338 / 0.000 / 0.007 / 56.9
CMS-CPCW/ H+SH+NGR [phiên (chuyển đạo)]	0.733 / 0.001 / 0.011 / 4.5	0.551 / 0.000 / 0.000 / 4.7	0.781 / 0.006 / 0.042 / 3.7	0.587 / 0.000 / 0.000 / 10.1	0.676 / 0.000 / 0.003 / 11.8	0.668 / 0.006 / 0.034 / 28.3	0.343 / 0.000 / 0.007 / 56.5
CMS-CPCW/ H+SH [phiên khác]	0.706 / 0.001 / 0.016 / 5.9	0.560 / 0.000 / 0.000 / 5.3	0.785 / 0.006 / 0.042 / 4.3	0.564 / 0.000 / 0.005 / 12.6	0.737 / 0.000 / 0.003 / 10.7	0.671 / 0.007 / 0.051 / 28.5	0.361 / 0.000 / 0.010 / 59.7
CMS-CPCW/ H+SH+NGRr [phiên khác]	0.706 / 0.001 / 0.011 / 5.2	0.559 / 0.000 / 0.000 / 4.6	0.785 / 0.006 / 0.042 / 3.6	0.561 / 0.000 / 0.003 / 12.9	0.736 / 0.000 / 0.003 / 10.8	0.661 / 0.006 / 0.040 / 29.3	0.352 / 0.000 / 0.007 / 59.5
CMS-CPCW/ H+SH+NGR [phiên khác]	0.706 / 0.001 / 0.011 / 5.4	0.559 / 0.000 / 0.000 / 4.8	0.785 / 0.006 / 0.042 / 3.7	0.563 / 0.000 / 0.003 / 12.6	0.736 / 0.000 / 0.003 / 10.8	0.664 / 0.006 / 0.040 / 29.0	0.356 / 0.000 / 0.007 / 59.6
CMS-CPCW/ H+SH [online nhân quả]	0.696 / 0.001 / 0.015 / 5.8	0.562 / 0.000 / 0.000 / 5.3	0.786 / 0.006 / 0.042 / 4.3	0.553 / 0.000 / 0.000 / 12.6	0.728 / 0.000 / 0.003 / 10.8	0.665 / 0.007 / 0.051 / 28.8	0.358 / 0.000 / 0.010 / 59.2
CMS-CPCW/ H+SH+NGRr [online nhân quả]	0.696 / 0.001 / 0.011 / 5.2	0.561 / 0.000 / 0.000 / 4.6	0.786 / 0.006 / 0.042 / 3.6	0.551 / 0.000 / 0.000 / 13.1	0.726 / 0.000 / 0.003 / 10.9	0.656 / 0.007 / 0.040 / 29.8	0.349 / 0.000 / 0.007 / 59.7
CMS-CPCW/ H+SH+NGR [online nhân quả]	0.696 / 0.001 / 0.011 / 5.4	0.561 / 0.000 / 0.000 / 4.8	0.786 / 0.006 / 0.042 / 3.7	0.552 / 0.000 / 0.000 / 12.8	0.727 / 0.000 / 0.003 / 10.9	0.659 / 0.007 / 0.040 / 29.4	0.353 / 0.000 / 0.007 / 59.5

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPCW/H+SH [giao phiên ∧ phiên khác]	0.693 / 0.001 / 0.015 / 5.2	0.534 / 0.000 / 0.000 / 5.9	0.778 / 0.006 / 0.042 / 4.1	0.500 / 0.000 / 0.000 / 13.3	0.636 / 0.000 / 0.003 / 12.5	0.655 / 0.006 / 0.045 / 29.6	0.288 / 0.000 / 0.010 / 68.7
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác]	0.693 / 0.000 / 0.011 / 5.0	0.534 / 0.000 / 0.000 / 5.1	0.778 / 0.006 / 0.042 / 3.6	0.497 / 0.000 / 0.000 / 13.8	0.635 / 0.000 / 0.003 / 12.5	0.645 / 0.005 / 0.040 / 30.5	0.279 / 0.000 / 0.007 / 68.9
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.693 / 0.000 / 0.011 / 5.1	0.534 / 0.000 / 0.000 / 5.3	0.778 / 0.006 / 0.042 / 3.7	0.499 / 0.000 / 0.000 / 13.4	0.635 / 0.000 / 0.003 / 12.5	0.649 / 0.005 / 0.040 / 30.0	0.283 / 0.000 / 0.007 / 69.0
CMS-CPC/H+SH+G [phiên (chuyển đạo)]	0.748 / 0.002 / 0.020 / 5.7	0.582 / 0.001 / 0.003 / 4.6	0.786 / 0.010 / 0.051 / 4.1	0.696 / 0.002 / 0.015 / 7.7	0.727 / 0.000 / 0.004 / 10.7	0.695 / 0.006 / 0.051 / 27.5	0.397 / 0.001 / 0.010 / 43.8
CMS-CPC/H+SH+G+NGRr [phiên (chuyển đạo)]	0.748 / 0.001 / 0.014 / 3.9	0.581 / 0.001 / 0.003 / 3.9	0.786 / 0.007 / 0.042 / 3.6	0.694 / 0.000 / 0.000 / 7.3	0.726 / 0.000 / 0.000 / 10.8	0.685 / 0.004 / 0.040 / 27.1	0.384 / 0.000 / 0.007 / 46.2
CMS-CPC/H+SH+G+NGR [phiên (chuyển đạo)]	0.748 / 0.001 / 0.014 / 4.1	0.581 / 0.001 / 0.003 / 3.9	0.786 / 0.007 / 0.042 / 3.6	0.694 / 0.000 / 0.000 / 7.2	0.726 / 0.000 / 0.000 / 10.8	0.689 / 0.004 / 0.040 / 26.7	0.389 / 0.000 / 0.007 / 45.6
CMS-CPC/H+SH+G [phiên khác]	0.726 / 0.002 / 0.020 / 5.7	0.576 / 0.002 / 0.010 / 5.9	0.787 / 0.010 / 0.051 / 4.2	0.660 / 0.002 / 0.010 / 8.9	0.758 / 0.000 / 0.004 / 10.3	0.691 / 0.007 / 0.051 / 27.3	0.410 / 0.002 / 0.026 / 54.7
CMS-CPC/H+SH+G+NGRr [phiên khác]	0.726 / 0.001 / 0.015 / 4.6	0.574 / 0.001 / 0.003 / 4.1	0.787 / 0.007 / 0.044 / 3.6	0.657 / 0.000 / 0.000 / 8.0	0.757 / 0.000 / 0.000 / 10.4	0.681 / 0.005 / 0.040 / 27.4	0.399 / 0.001 / 0.013 / 49.5
CMS-CPC/H+SH+G+NGR [phiên khác]	0.726 / 0.001 / 0.015 / 4.8	0.575 / 0.001 / 0.003 / 4.1	0.787 / 0.007 / 0.044 / 3.7	0.658 / 0.000 / 0.000 / 7.6	0.757 / 0.000 / 0.000 / 10.3	0.684 / 0.005 / 0.040 / 27.1	0.404 / 0.001 / 0.013 / 50.4
CMS-CPC/H+SH+G [online nhân quả]	0.702 / 0.001 / 0.017 / 5.2	0.542 / 0.000 / 0.000 / 4.9	0.778 / 0.006 / 0.042 / 4.0	0.539 / 0.000 / 0.000 / 10.8	0.729 / 0.000 / 0.003 / 10.8	0.666 / 0.005 / 0.051 / 28.7	0.356 / 0.000 / 0.010 / 52.5
CMS-CPC/H+SH+G+NGRr [online nhân quả]	0.702 / 0.001 / 0.012 / 4.8	0.541 / 0.000 / 0.000 / 4.0	0.778 / 0.006 / 0.042 / 3.5	0.536 / 0.000 / 0.000 / 11.1	0.728 / 0.000 / 0.000 / 10.8	0.656 / 0.004 / 0.040 / 29.8	0.345 / 0.000 / 0.007 / 53.5
CMS-CPC/H+SH+G+NGR [online nhân quả]	0.702 / 0.001 / 0.012 / 4.9	0.541 / 0.000 / 0.000 / 4.2	0.778 / 0.006 / 0.042 / 3.6	0.537 / 0.000 / 0.000 / 11.0	0.728 / 0.000 / 0.000 / 10.8	0.659 / 0.004 / 0.040 / 29.4	0.350 / 0.000 / 0.007 / 53.0
CMS-CPC/H+SH+G [giao phiên ∧ phiên khác]	0.713 / 0.001 / 0.019 / 5.0	0.567 / 0.001 / 0.003 / 4.9	0.782 / 0.009 / 0.051 / 4.1	0.629 / 0.001 / 0.008 / 8.6	0.688 / 0.000 / 0.004 / 11.4	0.680 / 0.005 / 0.051 / 27.8	0.339 / 0.001 / 0.010 / 56.5

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.713 / 0.001 / 0.013 / 4.6	0.566 / 0.000 / 0.000 / 4.2	0.782 / 0.007 / 0.044 / 3.5	0.627 / 0.000 / 0.000 / 9.1	0.687 / 0.000 / 0.000 / 11.4	0.669 / 0.004 / 0.040 / 28.5	0.326 / 0.000 / 0.007 / 57.1
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.713 / 0.001 / 0.013 / 4.8	0.566 / 0.000 / 0.000 / 4.3	0.782 / 0.007 / 0.044 / 3.6	0.628 / 0.000 / 0.000 / 8.9	0.687 / 0.000 / 0.000 / 11.4	0.673 / 0.004 / 0.040 / 28.1	0.332 / 0.000 / 0.007 / 57.4

Bảng 3. Độ chặt: tỷ lệ khởi phát sai mỗi đoạn / α và segFAR / α

Luật	α	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)]	0.5	0.03 / 0.04	0.07 / 0.02	0.08 / 0.18	0.05 / 0.01	0.08 / 0.01	0.09 / 0.14	0.08 / 0.01
CMS-CPCW/H+SH+NGR [phiên khác]	0.5	0.03 / 0.05	0.06 / 0.00	0.08 / 0.17	0.06 / 0.02	0.08 / 0.00	0.09 / 0.14	0.09 / 0.03
CMS-CPCW/H+SH+NGR [online nhân quả]	0.5	0.03 / 0.04	0.06 / 0.00	0.08 / 0.17	0.05 / 0.00	0.08 / 0.00	0.09 / 0.15	0.08 / 0.01
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.5	0.03 / 0.04	0.06 / 0.00	0.08 / 0.16	0.05 / 0.00	0.08 / 0.00	0.09 / 0.11	0.08 / 0.01
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)]	0.1	0.08 / 0.06	0.31 / 0.00	0.26 / 0.59	0.26 / 0.00	0.38 / 0.00	0.46 / 0.57	0.40 / 0.01
CMS-CPCW/H+SH+NGR [phiên khác]	0.1	0.08 / 0.07	0.31 / 0.00	0.26 / 0.59	0.26 / 0.00	0.38 / 0.00	0.47 / 0.65	0.40 / 0.02
CMS-CPCW/H+SH+NGR [online nhân quả]	0.1	0.08 / 0.06	0.31 / 0.00	0.26 / 0.58	0.26 / 0.00	0.38 / 0.00	0.46 / 0.68	0.40 / 0.02
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.1	0.08 / 0.05	0.31 / 0.00	0.26 / 0.57	0.26 / 0.00	0.38 / 0.00	0.47 / 0.53	0.40 / 0.01

Bảng 4. So sánh ghép cặp (cùng run): trung bình hiệu [CI 95 % bootstrap cụm theo seed] (p sign test theo seed)

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	Boreas	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	-0.2 [-0.8, +0.5] (p=3.4e-01)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NCLT	6	+0.021 [+0.008, +0.038] (p=1.0e+00)	-0.1 [-2.7, +3.3] (p=1.0e+00)	+0.001 [-0.002, +0.003]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NGRr vs CPC-SC-T0.5+SH	CMU	60	+0.037 [+0.028, +0.047] (p=2.0e-03)	-1.1 [-2.3, +0.0] (p=7.5e-01)	-0.009 [-0.019, +0.000]
CMS-CPCW/H+SH+NGRr vs CPC-SC-T0.5+SH	Nordland	60	+0.059 [+0.050, +0.066] (p=2.0e-03)	-3.7 [-6.1, -1.6] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+NGRr vs CPC-SC-T0.5+SH	MSR office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.8 [-2.1, -1.4] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CPC-SC-T0.5+SH	MSR old town	60	+0.061 [+0.052, +0.071] (p=2.0e-03)	-3.5 [-4.3, -2.7] (p=2.0e-03)	-0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+NGRr vs CPC-SC-T0.5+SH	MSR countryside	60	+0.134 [+0.111, +0.159] (p=2.0e-03)	-14.0 [-16.2, -12.2] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH[boreas]+NGRr vs CPC-SC-T0.5+SH	Boreas	60	+0.067 [+0.046, +0.083] (p=2.1e-02)	-0.8 [-1.6, -0.1] (p=1.1e-01)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH[NCLT]+NGRr vs CPC-SC-T0.5+SH	NCLT	6	+0.010 [+0.004, +0.018] (p=1.0e+00)	-1.2 [-2.7, -0.1] (p=1.0e+00)	-0.001 [-0.002, +0.000]
CMS-CPCW/H+SH[CMU]+NGRr vs CPC-SC-T0.5+SH	CMU	60	+0.026 [+0.017, +0.035] (p=2.0e-03)	-1.3 [-2.5, -0.2] (p=3.4e-01)	-0.010 [-0.020, -0.001]
CMS-CPCW/H+SH[Nordland]+NGRr vs CPC-SC-T0.5+SH	Nordland	60	+0.008 [-0.003, +0.018] (p=3.4e-01)	-2.2 [-3.6, -1.0] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPCW/H+SH[MSR office]+NGRr vs CPC-SC-T0.5+SH	MSR office	60	+0.028 [+0.019, +0.037] (p=2.0e-03)	-1.3 [-1.7, -1.0] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH[MSR old town]+NGRr vs CPC-SC-T0.5+SH	MSR old town	60	+0.043 [+0.034, +0.052] (p=2.0e-03)	-2.1 [-2.9, -1.3] (p=2.1e-02)	-0.001 [-0.003, -0.001]
CMS-CPCW/H+SH[MSR countryside]+NGRr vs CPC-SC-T0.5+SH	MSR countryside	60	+0.082 [+0.061, +0.106] (p=2.0e-03)	-8.9 [-10.8, -7.1] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH[boreas]+NGRr vs CPC-SC-T0.5+SH	Boreas	60	+0.062 [+0.041, +0.078] (p=2.1e-02)	-0.7 [-1.3, -0.2] (p=1.1e-01)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH[NCLT]+NGRr vs CPC-SC-T0.5+SH	NCLT	6	+0.015 [+0.006, +0.026] (p=1.0e+00)	-1.3 [-2.7, -0.1] (p=1.0e+00)	-0.001 [-0.002, +0.000]
CMS-CPCW/H+SH[CMU]+NGRr vs CPC-SC-T0.5+SH	CMU	60	+0.031 [+0.021, +0.040] (p=2.0e-03)	-1.2 [-2.4, -0.1] (p=7.5e-01)	-0.009 [-0.020, +0.000]
CMS-CPCW/H+SH[Nordland]+NGRr vs CPC-SC-T0.5+SH	Nordland	60	-0.019 [-0.029, -0.009] (p=2.1e-02)	-0.7 [-1.1, -0.4] (p=2.1e-02)	-0.001 [-0.001, +0.000]
CMS-CPCW/H+SH[MSR office]+NGRr vs CPC-SC-T0.5+SH	MSR office	60	+0.014 [+0.004, +0.024] (p=3.4e-01)	-1.2 [-1.5, -0.8] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH[MSR old town]+NGRr vs CPC-SC-T0.5+SH	MSR old town	60	+0.042 [+0.033, +0.052] (p=2.0e-03)	-1.9 [-2.7, -1.0] (p=2.1e-02)	+0.000 [-0.001, +0.001]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH[orig]+NGRr vs CPC-SC-T0.5+SH	4S-mc-Countryside	60	+0.123 [+0.099, +0.150] (p=2.0e-03)	-11.1 [-13.3, -8.9] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	BC-Open-Cross	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	-1.1 [-1.8, -0.7] (p=2.0e-03)	+0.001 [+0.000, +0.001]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S-Cross	6	+0.024 [-0.001, +0.052] (p=1.0e+00)	-0.6 [-2.8, +1.6] (p=1.0e+00)	+0.001 [-0.001, +0.004]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	CM-Cross	60	+0.033 [+0.023, +0.044] (p=2.0e-03)	-1.2 [-2.4, -0.1] (p=7.5e-01)	-0.008 [-0.018, +0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	BC-Orchard	60	+0.052 [+0.044, +0.059] (p=2.0e-03)	-3.9 [-6.2, -1.9] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S-Office	60	+0.061 [+0.053, +0.071] (p=2.0e-03)	-1.8 [-2.1, -1.4] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S-Old town	60	+0.063 [+0.054, +0.075] (p=2.0e-03)	-3.8 [-4.7, -2.9] (p=2.0e-03)	-0.001 [-0.003, +0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S-Countryside	60	+0.156 [+0.134, +0.180] (p=2.0e-03)	-17.0 [-18.9, -15.1] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	BC-Cross	60	+0.085 [+0.064, +0.100] (p=2.0e-03)	-1.4 [-2.1, -0.8] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	4S-Cross	6	+0.015 [-0.003, +0.037] (p=1.0e+00)	-0.6 [-2.8, +1.6] (p=1.0e+00)	+0.001 [-0.002, +0.003]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	CM-Cross	60	+0.023 [+0.014, +0.033] (p=2.1e-02)	-1.4 [-2.6, -0.3] (p=3.4e-01)	-0.009 [-0.020, -0.000]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	BC-Orchard	60	+0.021 [+0.012, +0.028] (p=2.1e-02)	-3.2 [-5.0, -1.6] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	4S-Office	60	+0.036 [+0.028, +0.045] (p=2.0e-03)	-1.5 [-1.9, -1.1] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	4S-Old town	60	+0.052 [+0.042, +0.063] (p=2.0e-03)	-3.0 [-3.8, -2.2] (p=2.0e-03)	-0.002 [-0.003, -0.001]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	4S-Countryside	60	+0.115 [+0.095, +0.137] (p=2.0e-03)	-13.1 [-15.1, -11.1] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	Bonus	60	+0.074 [+0.053, +0.089] (p=2.1e-02)	-1.2 [-1.9, -0.6] (p=2.1e-02)	+0.000 [-0.000, +0.001]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	4S-Cross	6	+0.010 [-0.006, +0.029] (p=1.0e+00)	-0.8 [-2.9, +1.5] (p=1.0e+00)	+0.000 [-0.002, +0.002]
CMS-CPC/H+SH+G[bonus]+NGRr vs CPC-SC-T0.5+SH	CM-Cross	60	+0.023 [+0.014, +0.033] (p=2.1e-02)	-1.3 [-2.5, -0.2] (p=3.4e-01)	-0.009 [-0.019, -0.000]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	Nordic	60	-0.031 [-0.039, -0.024] (p=2.0e-03)	-0.8 [-1.2, -0.3] (p=1.1e-01)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S office	60	+0.018 [+0.008, +0.028] (p=3.4e-01)	-1.2 [-1.6, -0.8] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S old town	60	+0.040 [+0.031, +0.051] (p=2.0e-03)	-1.7 [-2.5, -0.9] (p=2.1e-02)	-0.002 [-0.003, -0.000]
CMS-CPC/H+SH+G+NGRr vs CPC-SC-T0.5+SH	4S countryside	60	+0.131 [+0.106, +0.157] (p=2.0e-03)	-13.7 [-15.8, -11.7] (p=2.0e-03)	+0.000 [-0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	BGR	60	-0.000 [-0.000, +0.000] (p=7.3e-01)	+2.9 [+1.8, +4.1] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	NGLT	6	+0.000 [+0.000, +0.001] (p=1.0e+00)	-0.3 [-1.0, +0.1] (p=1.0e+00)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	CMU	60	-0.000 [-0.000, -0.000] (p=2.2e-01)	+0.6 [+0.3, +0.9] (p=2.0e-03)	+0.000 [+0.000, +0.001]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	Nordland	60	+0.000 [+0.000, +0.001] (p=1.2e-01)	+0.0 [-0.2, +0.3] (p=3.8e-01)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	4S office	60	+0.000 [+0.000, +0.000] (p=7.8e-03)	+0.0 [-0.0, +0.1] (p=2.9e-01)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	4S old town	60	+0.003 [+0.003, +0.004] (p=2.0e-03)	+0.4 [-0.3, +1.5] (p=3.4e-01)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGRr vs CMS-CPCW/H+SH+NGRr	4S countryside	60	+0.004 [+0.003, +0.004] (p=2.0e-03)	-0.5 [-1.0, +0.1] (p=1.1e-01)	+0.000 [+0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	BGR	60	-0.000 [-0.000, -0.000] (p=3.1e-02)	+2.0 [+1.4, +2.8] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	NGLT	6	+0.000 [+0.000, +0.001] (p=1.0e+00)	+0.6 [+0.0, +1.7] (p=1.0e+00)	+0.001 [+0.000, +0.002]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	CMU	60	-0.000 [-0.000, -0.000] (p=6.2e-02)	+0.5 [+0.3, +0.8] (p=2.0e-03)	+0.000 [+0.000, +0.001]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	Nordland	60	+0.000 [+0.000, +0.001] (p=2.5e-01)	-0.1 [-0.4, +0.0] (p=2.5e-01)	+0.000 [+0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	4S office	60	+0.000 [+0.000, +0.000] (p=7.8e-03)	+0.0 [-0.0, +0.0] (p=2.9e-01)	+0.000 [+0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	4S old town	60	+0.003 [+0.003, +0.004] (p=2.0e-03)	-0.3 [-0.4, -0.2] (p=2.1e-02)	+0.000 [+0.000, +0.000]
CMS-CPC/H+SH+G+NGRr vs CMS-CPC/H+SH+G+NGRr	4S countryside	60	+0.003 [+0.002, +0.005] (p=2.0e-03)	-1.0 [-1.4, -0.6] (p=2.0e-03)	+0.000 [+0.000, +0.000]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH[boreas] vs CMS-CPCW/H+SH+NGRr	Boreas	60	-0.033 [-0.035, -0.031] (p=2.0e-03)	-0.6 [-1.3, +0.0] (p=7.5e-01)	-0.000 [-0.000, -0.000]
CMS-CPCW/H+SH[nclt] vs CMS-CPCW/H+SH+NGRr	NCLT	6	-0.011 [-0.021, -0.003] (p=1.0e+00)	-1.2 [-3.6, +0.1] (p=1.0e+00)	-0.001 [-0.003, +0.000]
CMS-CPCW/H+SH[cmu] vs CMS-CPCW/H+SH+NGRr	CMU	60	-0.011 [-0.013, -0.010] (p=2.0e-03)	-0.2 [-0.3, -0.2] (p=2.0e-03)	-0.001 [-0.002, -0.000]
CMS-CPCW/H+SH[nordland] vs CMS-CPCW/H+SH+NGRr	Nordland	60	-0.051 [-0.056, -0.046] (p=2.0e-03)	+1.5 [+0.5, +2.8] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH[4s office] vs CMS-CPCW/H+SH+NGRr	4S office	60	-0.032 [-0.034, -0.030] (p=2.0e-03)	+0.4 [+0.3, +0.5] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH[4s old town] vs CMS-CPCW/H+SH+NGRr	4S old town	60	-0.018 [-0.020, -0.016] (p=2.0e-03)	+1.4 [+1.2, +1.6] (p=2.0e-03)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH[4s countryside] vs CMS-CPCW/H+SH+NGRr	4S countryside	60	-0.051 [-0.057, -0.046] (p=2.0e-03)	+5.1 [+4.3, +6.0] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	Boreas	60	+0.067 [+0.048, +0.084] (p=2.1e-02)	-1.3 [-2.0, -0.8] (p=2.0e-03)	-0.001 [-0.003, -0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	NCLT	6	-0.038 [-0.069, -0.017] (p=1.0e+00)	-0.4 [-2.0, +1.2] (p=1.0e+00)	-0.001 [-0.002, +0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	CMU	60	-0.008 [-0.016, +0.000] (p=3.4e-01)	-1.6 [-2.8, -0.5] (p=2.1e-02)	-0.012 [-0.022, -0.003]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	Nordland	60	-0.127 [-0.152, -0.103] (p=2.0e-03)	+0.7 [-0.4, +1.7] (p=3.4e-01)	-0.001 [-0.001, -0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	4S office	60	-0.063 [-0.187, +0.025] (p=1.1e-01)	+0.3 [-1.4, +2.5] (p=1.1e-01)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	4S old town	60	+0.025 [+0.014, +0.038] (p=2.0e-03)	-1.1 [-2.1, -0.2] (p=3.4e-01)	-0.001 [-0.003, -0.000]
CMS-CPCW/H+SH(a=0.01)+NGRr vs CPC-SC-T0.5+SH	4S countryside	60	+0.004 [-0.020, +0.029] (p=7.5e-01)	+5.6 [+2.4, +8.1] (p=2.1e-02)	-0.000 [-0.001, +0.000]

Bảng 5a. Độ bền bộ lọc — F1: nhiễu quá trình 0.2·L (×4), α = 0.05

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CPC-SC+CS	0.661 / 0.001 / 0.021 / 6.0	0.537 / 0.000 / 0.000 / 4.3	0.731 / 0.002 / 0.019 / 3.8	0.644 / 0.000 / 0.003 / 7.5	0.709 / 0.000 / 0.000 / 9.0	0.600 / 0.000 / 0.010 / 24.4	0.297 / 0.000 / 0.007 / 48.0
CPC-SC-T½+SH	0.666 / 0.002 / 0.023 / 7.5	0.588 / 0.001 / 0.003 / 4.8	0.789 / 0.018 / 0.188 / 6.1	0.713 / 0.001 / 0.010 / 8.1	0.738 / 0.000 / 0.003 / 9.8	0.640 / 0.007 / 0.034 / 24.7	0.334 / 0.000 / 0.015 / 46.1

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CPC-SC-T½+SH+NGRr	0.766 / 0.001 / 0.016 / 6.8	0.588 / 0.000 / 0.000 / 3.8	0.789 / 0.011 / 0.109 / 4.3	0.713 / 0.000 / 0.003 / 7.5	0.738 / 0.000 / 0.003 / 9.8	0.630 / 0.006 / 0.034 / 25.3	0.328 / 0.000 / 0.003 / 47.0
CPC-SC-T½+SH+NGR	0.766 / 0.001 / 0.017 / 6.9	0.588 / 0.000 / 0.000 / 3.9	0.789 / 0.012 / 0.129 / 5.1	0.713 / 0.000 / 0.003 / 7.5	0.738 / 0.000 / 0.003 / 9.8	0.634 / 0.006 / 0.034 / 25.0	0.330 / 0.000 / 0.004 / 47.0
CMS-CPCW/ H+SH	0.766 / 0.004 / 0.028 / 12.7	0.611 / 0.005 / 0.029 / 5.0	0.827 / 0.013 / 0.065 / 7.7	0.773 / 0.002 / 0.018 / 12.1	0.799 / 0.001 / 0.015 / 8.8	0.710 / 0.008 / 0.051 / 24.0	0.479 / 0.001 / 0.010 / 31.7
CMS-CPCW/ H+SH+NGRr	0.765 / 0.002 / 0.020 / 8.6	0.610 / 0.001 / 0.007 / 4.3	0.826 / 0.009 / 0.054 / 4.5	0.773 / 0.000 / 0.010 / 3.9	0.798 / 0.000 / 0.003 / 8.2	0.701 / 0.007 / 0.040 / 21.8	0.472 / 0.000 / 0.007 / 31.3
CMS-CPCW/ H+SH+NGR	0.765 / 0.002 / 0.022 / 13.2	0.610 / 0.001 / 0.007 / 4.1	0.826 / 0.009 / 0.056 / 5.5	0.773 / 0.000 / 0.010 / 4.5	0.798 / 0.000 / 0.006 / 8.3	0.704 / 0.007 / 0.040 / 22.2	0.473 / 0.000 / 0.007 / 31.6
CMS-CPC/ H+SH+G	0.776 / 0.005 / 0.030 / 17.4	0.614 / 0.008 / 0.042 / 5.9	0.823 / 0.017 / 0.067 / 7.8	0.767 / 0.004 / 0.020 / 12.0	0.801 / 0.002 / 0.018 / 8.5	0.713 / 0.008 / 0.062 / 25.2	0.503 / 0.002 / 0.026 / 29.6
CMS-CPC/ H+SH+G+NGRr	0.775 / 0.002 / 0.020 / 7.0	0.612 / 0.002 / 0.007 / 4.0	0.822 / 0.009 / 0.052 / 4.3	0.766 / 0.000 / 0.000 / 3.0	0.800 / 0.000 / 0.003 / 7.7	0.704 / 0.006 / 0.051 / 20.6	0.496 / 0.001 / 0.010 / 27.0
CMS-CPC/ H+SH+G+NGR	0.775 / 0.003 / 0.022 / 10.5	0.612 / 0.002 / 0.010 / 4.2	0.822 / 0.010 / 0.052 / 5.1	0.766 / 0.000 / 0.000 / 3.0	0.800 / 0.000 / 0.006 / 7.8	0.708 / 0.006 / 0.051 / 20.6	0.497 / 0.001 / 0.010 / 27.9
CMS-CPC/ H+SH	0.781 / 0.006 / 0.030 / 15.2	0.630 / 0.013 / 0.059 / 8.2	0.834 / 0.018 / 0.072 / 8.6	0.798 / 0.009 / 0.044 / 24.1	0.806 / 0.002 / 0.018 / 8.9	0.719 / 0.011 / 0.062 / 26.4	0.523 / 0.003 / 0.026 / 31.8
CMS-CPC/ H+SH+NGRr	0.780 / 0.003 / 0.020 / 9.5	0.627 / 0.003 / 0.016 / 4.7	0.833 / 0.010 / 0.058 / 4.7	0.796 / 0.001 / 0.013 / 5.2	0.805 / 0.000 / 0.003 / 8.1	0.709 / 0.008 / 0.051 / 22.3	0.516 / 0.001 / 0.013 / 27.5
CMS-CPC/ H+SH+NGR	0.780 / 0.003 / 0.022 / 14.8	0.628 / 0.004 / 0.023 / 4.5	0.833 / 0.011 / 0.062 / 5.9	0.797 / 0.002 / 0.015 / 7.2	0.805 / 0.000 / 0.006 / 8.1	0.713 / 0.008 / 0.051 / 22.4	0.518 / 0.001 / 0.016 / 31.2
ablation +NG	0.781 / 0.001 / 0.009 / 5.4	0.617 / 0.001 / 0.003 / 4.9	0.833 / 0.008 / 0.054 / 3.8	0.767 / 0.000 / 0.003 / 7.1	0.792 / 0.000 / 0.003 / 8.7	0.518 / 0.005 / 0.032 / 71.7	0.344 / 0.000 / 0.004 / 64.2
ablation +DRG	0.771 / 0.001 / 0.013 / 25.1	0.557 / 0.006 / 0.029 / 53.8	0.826 / 0.015 / 0.067 / 9.3	0.769 / 0.000 / 0.005 / 11.8	0.805 / 0.000 / 0.003 / 8.0	0.566 / 0.007 / 0.051 / 70.3	0.493 / 0.001 / 0.013 / 36.2
Xu'21 (nhãn)	0.684 / 0.043 / 0.567 / 34.6	0.526 / 0.021 / 0.065 / 12.7	0.686 / 0.227 / 1.000 / 23.3	0.740 / 0.038 / 0.113 / 80.3	0.724 / 0.215 / 0.997 / 24.3	0.625 / 0.101 / 1.000 / 84.5	0.254 / 0.088 / 0.983 / 84.4
Claxton'24 (nhãn)	0.741 / 0.116 / 0.674 / 57.3	0.660 / 0.203 / 0.428 / 33.3	0.874 / 0.167 / 0.818 / 23.8	0.520 / 0.262 / 0.442 / 169.0	0.799 / 0.156 / 0.589 / 27.3	0.724 / 0.165 / 0.544 / 120.4	0.598 / 0.215 / 0.493 / 60.0
TAO'26 (nhãn)	0.801 / 0.240 / 0.830 / 53.2	0.756 / 0.707 / 1.000 / 57.2	0.930 / 0.370 / 1.000 / 45.5	0.805 / 0.579 / 1.000 / 214.6	0.935 / 0.740 / 1.000 / 62.4	0.839 / 0.557 / 1.000 / 185.8	0.769 / 0.732 / 1.000 / 108.5

Bảng 5b. Độ bền bộ lọc — F2: F1 + σ fix 10 m, α = 0.05

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CPC-SC+CS	0.661 / 0.001 / 0.021 / 5.3	0.537 / 0.000 / 0.000 / 4.0	0.731 / 0.002 / 0.019 / 3.3	0.644 / 0.000 / 0.003 / 8.2	0.709 / 0.000 / 0.000 / 9.9	0.600 / 0.000 / 0.010 / 26.3	0.297 / 0.000 / 0.007 / 49.5
CPC-SC-T½+SH	0.666 / 0.002 / 0.023 / 6.4	0.588 / 0.001 / 0.003 / 4.2	0.789 / 0.018 / 0.188 / 4.6	0.713 / 0.001 / 0.010 / 8.4	0.738 / 0.000 / 0.003 / 10.0	0.640 / 0.007 / 0.034 / 25.9	0.334 / 0.000 / 0.015 / 47.6
CPC-SC-T½+SH+NGRr	0.666 / 0.001 / 0.015 / 5.7	0.588 / 0.000 / 0.000 / 3.5	0.789 / 0.011 / 0.113 / 3.7	0.713 / 0.000 / 0.003 / 8.2	0.738 / 0.000 / 0.003 / 10.0	0.628 / 0.006 / 0.034 / 27.1	0.326 / 0.000 / 0.004 / 48.7
CPC-SC-T½+SH+NGR	0.666 / 0.001 / 0.017 / 5.9	0.588 / 0.000 / 0.000 / 3.5	0.789 / 0.011 / 0.125 / 4.0	0.713 / 0.000 / 0.003 / 8.1	0.738 / 0.000 / 0.003 / 10.0	0.634 / 0.006 / 0.034 / 26.4	0.329 / 0.000 / 0.004 / 48.6
CMS-CPCW/ H+SH	0.766 / 0.004 / 0.028 / 9.1	0.611 / 0.005 / 0.029 / 4.5	0.827 / 0.013 / 0.065 / 5.1	0.773 / 0.002 / 0.018 / 9.2	0.799 / 0.001 / 0.015 / 8.6	0.710 / 0.008 / 0.051 / 24.0	0.479 / 0.001 / 0.010 / 32.1
CMS-CPCW/ H+SH+NGRr	0.765 / 0.002 / 0.020 / 6.4	0.611 / 0.001 / 0.007 / 3.4	0.826 / 0.009 / 0.052 / 3.8	0.772 / 0.000 / 0.010 / 4.4	0.798 / 0.000 / 0.003 / 8.2	0.698 / 0.007 / 0.040 / 22.7	0.469 / 0.000 / 0.007 / 32.7
CMS-CPCW/ H+SH+NGR	0.765 / 0.002 / 0.022 / 8.2	0.610 / 0.001 / 0.007 / 4.0	0.826 / 0.009 / 0.054 / 4.1	0.773 / 0.000 / 0.010 / 4.6	0.798 / 0.000 / 0.006 / 8.3	0.704 / 0.007 / 0.040 / 23.1	0.473 / 0.000 / 0.007 / 32.8
CMS-CPC/ H+SH+G	0.776 / 0.005 / 0.030 / 11.1	0.614 / 0.008 / 0.042 / 5.0	0.823 / 0.017 / 0.067 / 5.1	0.767 / 0.004 / 0.020 / 7.5	0.801 / 0.002 / 0.018 / 8.5	0.713 / 0.008 / 0.062 / 24.3	0.503 / 0.002 / 0.026 / 30.2
CMS-CPC/ H+SH+G+NGRr	0.775 / 0.002 / 0.021 / 5.1	0.613 / 0.002 / 0.007 / 3.3	0.822 / 0.009 / 0.050 / 3.6	0.766 / 0.000 / 0.000 / 3.9	0.800 / 0.000 / 0.003 / 8.0	0.702 / 0.006 / 0.051 / 21.9	0.493 / 0.001 / 0.010 / 29.1
CMS-CPC/ H+SH+G+NGR	0.775 / 0.003 / 0.022 / 6.4	0.613 / 0.002 / 0.010 / 3.8	0.822 / 0.010 / 0.050 / 3.9	0.766 / 0.000 / 0.000 / 3.9	0.800 / 0.000 / 0.006 / 8.1	0.707 / 0.006 / 0.051 / 21.7	0.496 / 0.001 / 0.010 / 28.6
CMS-CPC/ H+SH	0.781 / 0.006 / 0.030 / 11.4	0.630 / 0.013 / 0.059 / 6.3	0.834 / 0.018 / 0.072 / 5.5	0.798 / 0.009 / 0.044 / 16.3	0.806 / 0.002 / 0.018 / 8.6	0.719 / 0.011 / 0.062 / 25.5	0.523 / 0.003 / 0.026 / 31.5
CMS-CPC/ H+SH+NGRr	0.780 / 0.003 / 0.021 / 6.7	0.628 / 0.004 / 0.020 / 3.6	0.833 / 0.010 / 0.058 / 3.9	0.796 / 0.001 / 0.013 / 4.6	0.805 / 0.000 / 0.003 / 8.1	0.707 / 0.008 / 0.051 / 22.8	0.514 / 0.001 / 0.013 / 27.8
CMS-CPC/ H+SH+NGR	0.780 / 0.003 / 0.022 / 9.5	0.628 / 0.004 / 0.023 / 4.1	0.833 / 0.010 / 0.058 / 4.2	0.797 / 0.002 / 0.023 / 6.5	0.805 / 0.000 / 0.006 / 8.2	0.713 / 0.008 / 0.051 / 23.0	0.517 / 0.001 / 0.016 / 31.0
ablation +NG	0.781 / 0.001 / 0.007 / 4.1	0.628 / 0.001 / 0.003 / 3.1	0.833 / 0.007 / 0.040 / 3.3	0.765 / 0.000 / 0.003 / 8.1	0.789 / 0.000 / 0.003 / 8.8	0.517 / 0.004 / 0.028 / 71.7	0.337 / 0.000 / 0.004 / 66.1
ablation +DRG	0.771 / 0.001 / 0.013 / 12.7	0.557 / 0.006 / 0.029 / 38.1	0.826 / 0.015 / 0.067 / 5.7	0.769 / 0.000 / 0.005 / 6.1	0.805 / 0.000 / 0.003 / 8.0	0.566 / 0.007 / 0.051 / 67.6	0.493 / 0.001 / 0.013 / 37.7
Xu'21 (nhãn)	0.684 / 0.043 / 0.567 / 30.0	0.526 / 0.021 / 0.065 / 11.5	0.686 / 0.227 / 1.000 / 17.8	0.740 / 0.038 / 0.113 / 66.7	0.724 / 0.215 / 0.997 / 22.3	0.625 / 0.101 / 1.000 / 78.1	0.254 / 0.088 / 0.983 / 82.2
Claxton'24 (nhãn)	0.741 / 0.116 / 0.674 / 46.6	0.660 / 0.203 / 0.428 / 27.4	0.874 / 0.167 / 0.818 / 14.0	0.520 / 0.262 / 0.442 / 120.3	0.799 / 0.156 / 0.589 / 22.2	0.724 / 0.165 / 0.544 / 102.1	0.598 / 0.215 / 0.493 / 49.0

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
TAO'26 (nhãn)	0.801 / 0.240 / 0.830 / 49.7	0.756 / 0.707 / 1.000 / 56.0	0.930 / 0.370 / 1.000 / 35.5	0.805 / 0.579 / 1.000 / 200.1	0.935 / 0.740 / 1.000 / 58.0	0.839 / 0.557 / 1.000 / 175.5	0.769 / 0.732 / 1.000 / 102.1

Bảng 5c. Độ bền bộ lọc — F3: khởi tạo không nhãn (top-1 frame đầu, σ 1 km), α = 0.05

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
CPC-SC+CS	0.661 / 0.001 / 0.021 / 40.4	0.537 / 0.000 / 0.000 / 22.7	0.731 / 0.002 / 0.019 / 31.9	0.644 / 0.000 / 0.003 / 96.6	0.709 / 0.000 / 0.000 / 11.6	0.600 / 0.000 / 0.010 / 113.2	0.297 / 0.000 / 0.007 / 54.6
CPC-SC-T½+SH	0.666 / 0.002 / 0.023 / 43.1	0.588 / 0.001 / 0.003 / 20.3	0.789 / 0.018 / 0.188 / 26.2	0.713 / 0.001 / 0.010 / 96.0	0.738 / 0.000 / 0.003 / 11.4	0.640 / 0.007 / 0.034 / 115.0	0.334 / 0.000 / 0.015 / 52.9
CPC-SC-T½+SH+NGRr	0.666 / 0.001 / 0.017 / 42.7	0.587 / 0.000 / 0.000 / 20.5	0.788 / 0.015 / 0.167 / 26.3	0.712 / 0.000 / 0.003 / 98.2	0.738 / 0.000 / 0.003 / 11.5	0.629 / 0.006 / 0.034 / 116.6	0.326 / 0.000 / 0.004 / 54.0
CPC-SC-T½+SH+NGR	0.666 / 0.001 / 0.017 / 42.8	0.587 / 0.000 / 0.000 / 20.3	0.788 / 0.015 / 0.181 / 26.1	0.712 / 0.000 / 0.003 / 98.1	0.738 / 0.000 / 0.003 / 11.5	0.633 / 0.006 / 0.034 / 115.8	0.328 / 0.000 / 0.004 / 53.8
CMS-CPCW/ H+SH	0.766 / 0.004 / 0.028 / 19.2	0.611 / 0.005 / 0.029 / 17.3	0.827 / 0.013 / 0.065 / 23.6	0.773 / 0.002 / 0.018 / 12.4	0.799 / 0.001 / 0.015 / 9.8	0.710 / 0.008 / 0.051 / 64.6	0.479 / 0.001 / 0.010 / 35.6
CMS-CPCW/ H+SH+NGRr	0.765 / 0.002 / 0.022 / 25.1	0.608 / 0.003 / 0.016 / 17.5	0.824 / 0.012 / 0.062 / 24.5	0.770 / 0.000 / 0.013 / 12.6	0.798 / 0.000 / 0.006 / 9.7	0.700 / 0.008 / 0.051 / 64.2	0.468 / 0.000 / 0.007 / 37.1
CMS-CPCW/ H+SH+NGR	0.766 / 0.002 / 0.022 / 24.6	0.610 / 0.003 / 0.016 / 16.9	0.825 / 0.012 / 0.062 / 24.0	0.772 / 0.000 / 0.013 / 12.3	0.798 / 0.000 / 0.006 / 9.7	0.704 / 0.008 / 0.051 / 64.6	0.472 / 0.000 / 0.007 / 36.7
CMS-CPC/ H+SH+G	0.776 / 0.005 / 0.030 / 21.8	0.614 / 0.008 / 0.042 / 14.9	0.823 / 0.017 / 0.067 / 23.5	0.767 / 0.004 / 0.020 / 10.9	0.801 / 0.002 / 0.018 / 9.9	0.713 / 0.008 / 0.062 / 63.5	0.503 / 0.002 / 0.026 / 33.9
CMS-CPC/ H+SH+G+NGRr	0.775 / 0.003 / 0.022 / 24.8	0.610 / 0.005 / 0.029 / 15.4	0.819 / 0.014 / 0.062 / 24.4	0.761 / 0.000 / 0.005 / 14.5	0.800 / 0.000 / 0.006 / 9.7	0.702 / 0.006 / 0.062 / 63.2	0.491 / 0.001 / 0.010 / 34.4
CMS-CPC/ H+SH+G+NGR	0.775 / 0.003 / 0.022 / 23.4	0.611 / 0.006 / 0.033 / 14.8	0.821 / 0.014 / 0.065 / 23.7	0.764 / 0.000 / 0.005 / 12.9	0.800 / 0.000 / 0.006 / 9.7	0.706 / 0.006 / 0.062 / 62.8	0.494 / 0.001 / 0.010 / 33.4
CMS-CPC/ H+SH	0.781 / 0.006 / 0.030 / 18.5	0.630 / 0.013 / 0.059 / 15.7	0.834 / 0.018 / 0.072 / 23.6	0.798 / 0.009 / 0.044 / 16.1	0.806 / 0.002 / 0.018 / 9.8	0.719 / 0.011 / 0.062 / 64.0	0.523 / 0.003 / 0.026 / 34.3
CMS-CPC/ H+SH+NGRr	0.780 / 0.003 / 0.022 / 24.2	0.624 / 0.007 / 0.036 / 14.4	0.830 / 0.015 / 0.071 / 24.5	0.792 / 0.002 / 0.018 / 13.4	0.804 / 0.000 / 0.006 / 9.6	0.708 / 0.009 / 0.062 / 62.9	0.513 / 0.001 / 0.013 / 31.9
CMS-CPC/ H+SH+NGR	0.780 / 0.003 / 0.022 / 23.3	0.627 / 0.008 / 0.042 / 13.7	0.832 / 0.015 / 0.071 / 23.9	0.796 / 0.002 / 0.018 / 13.3	0.804 / 0.000 / 0.006 / 9.6	0.712 / 0.009 / 0.062 / 63.3	0.516 / 0.001 / 0.016 / 33.8

Luật	Boreas	NCLT	CMU	Nordland	4S office	4S old town	4S countryside
ablation +NG	0.772 / 0.001 / 0.009 / 31.5	0.589 / 0.005 / 0.026 / 51.7	0.824 / 0.014 / 0.065 / 26.1	0.577 / 0.001 / 0.010 / 864.5	0.786 / 0.000 / 0.003 / 10.4	0.456 / 0.005 / 0.034 / 128.6	0.314 / 0.000 / 0.004 / 72.5
ablation +DRG	0.771 / 0.001 / 0.013 / 39.8	0.557 / 0.006 / 0.029 / 53.6	0.826 / 0.015 / 0.067 / 25.8	0.769 / 0.000 / 0.005 / 68.7	0.805 / 0.000 / 0.003 / 9.5	0.566 / 0.007 / 0.051 / 116.3	0.493 / 0.001 / 0.013 / 42.0
Xu'21 (nhãn)	0.684 / 0.043 / 0.567 / 43.3	0.526 / 0.021 / 0.065 / 23.4	0.686 / 0.227 / 1.000 / 19.4	0.740 / 0.038 / 0.113 / 57.3	0.724 / 0.215 / 0.997 / 21.2	0.625 / 0.101 / 1.000 / 131.5	0.254 / 0.088 / 0.983 / 81.5
Claxton'24 (nhãn)	0.741 / 0.116 / 0.619 / 39.1	0.663 / 0.213 / 0.428 / 30.5	0.874 / 0.166 / 0.809 / 13.2	0.521 / 0.266 / 0.447 / 89.3	0.798 / 0.152 / 0.589 / 19.0	0.724 / 0.167 / 0.544 / 94.1	0.597 / 0.213 / 0.498 / 42.7
TAO'26 (nhãn)	0.801 / 0.240 / 0.830 / 65.8	0.756 / 0.707 / 1.000 / 49.3	0.930 / 0.370 / 1.000 / 19.2	0.805 / 0.579 / 1.000 / 181.5	0.935 / 0.740 / 1.000 / 54.0	0.839 / 0.557 / 1.000 / 164.1	0.769 / 0.732 / 1.000 / 95.3

Bảng 6. Ghép cặp dưới F1–F3

f1

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	Boreas	60	+0.099 [+0.079, +0.115] (p=2.0e-03)	+1.1 [-0.3, +2.3] (p=3.4e-01)	+0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	NCLT	6	+0.021 [+0.008, +0.037] (p=1.0e+00)	-0.5 [-2.6, +1.6] (p=1.0e+00)	+0.001 [-0.002, +0.003]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	CMU	60	+0.037 [+0.028, +0.047] (p=2.0e-03)	-1.6 [-3.5, +0.1] (p=7.5e-01)	-0.009 [-0.019, -0.000]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	Nordland	60	+0.059 [+0.051, +0.066] (p=2.0e-03)	-4.2 [-7.3, -1.3] (p=2.1e-02)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	4S office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.6 [-2.1, -1.2] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	4S old town	60	+0.061 [+0.052, +0.072] (p=2.0e-03)	-2.9 [-4.0, -1.7] (p=2.1e-02)	+0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+NG vs CPC-SC-T0.5+SH	4S countryside	60	+0.137 [+0.115, +0.162] (p=2.0e-03)	-14.9 [-17.2, -12.6] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GNG vs CPC-SC-T0.5+SH	Boreas	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	-0.5 [-1.9, +0.6] (p=1.0e+00)	+0.000 [-0.000, +0.001]
CMS-CPC/H+SH+GNG vs CPC-SC-T0.5+SH	NCLT	6	+0.024 [-0.001, +0.053] (p=1.0e+00)	-0.8 [-2.8, +0.8] (p=1.0e+00)	+0.001 [-0.001, +0.004]
CMS-CPC/H+SH+GNG vs CPC-SC-T0.5+SH	CMU	60	+0.033 [+0.023, +0.043] (p=2.0e-03)	-1.8 [-3.7, -0.1] (p=7.5e-01)	-0.008 [-0.018, +0.000]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL woodland	60	+0.053 [+0.045, +0.060] (p=2.0e-03)	-5.1 [-8.3, -2.2] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL office	60	+0.062 [+0.053, +0.071] (p=2.0e-03)	-2.1 [-2.6, -1.7] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL old town	60	+0.065 [+0.054, +0.077] (p=2.0e-03)	-4.1 [-5.3, -2.8] (p=2.1e-02)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL countryside	60	+0.161 [+0.139, +0.185] (p=2.0e-03)	-19.1 [-21.6, -16.3] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL Korea	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	+5.6 [+3.3, +8.1] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL UT	6	+0.021 [+0.008, +0.038] (p=1.0e+00)	-0.7 [-2.6, +0.8] (p=1.0e+00)	+0.001 [-0.002, +0.003]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL CMU	60	+0.037 [+0.028, +0.047] (p=2.0e-03)	-0.6 [-2.7, +1.4] (p=7.5e-01)	-0.008 [-0.019, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL woodland	60	+0.059 [+0.051, +0.067] (p=2.0e-03)	-3.6 [-7.1, +0.1] (p=2.1e-02)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL office	60	+0.060 [+0.050, +0.069] (p=2.0e-03)	-1.6 [-2.1, -1.1] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL old town	60	+0.064 [+0.056, +0.075] (p=2.0e-03)	-2.4 [-4.0, -0.6] (p=1.1e-01)	+0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGRL countryside	60	+0.139 [+0.116, +0.164] (p=2.0e-03)	-14.5 [-17.1, -11.9] (p=2.0e-03)	+0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL Korea	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	+3.0 [+1.5, +4.6] (p=1.1e-01)	+0.001 [+0.000, +0.002]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL UT	6	+0.024 [-0.001, +0.053] (p=1.0e+00)	-0.6 [-2.8, +1.2] (p=1.0e+00)	+0.002 [-0.001, +0.005]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL CMU	60	+0.033 [+0.023, +0.043] (p=2.0e-03)	-0.9 [-3.0, +1.1] (p=7.5e-01)	-0.008 [-0.018, +0.001]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL woodland	60	+0.053 [+0.045, +0.060] (p=2.0e-03)	-5.1 [-8.3, -2.2] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL office	60	+0.062 [+0.053, +0.071] (p=2.0e-03)	-2.0 [-2.6, -1.6] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL old town	60	+0.068 [+0.058, +0.080] (p=2.0e-03)	-4.1 [-5.4, -2.7] (p=2.1e-02)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+G+NGR vs CPC-SC-T0.5+SH	NGRL countryside	60	+0.163 [+0.141, +0.187] (p=2.0e-03)	-18.2 [-21.6, -13.8] (p=2.0e-03)	+0.000 [-0.000, +0.001]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR_area	60	-0.001 [-0.001, -0.000] (p=2.0e-03)	-4.1 [-6.0, -2.4] (p=2.0e-03)	-0.002 [-0.003, -0.001]
CMS-CPCW/H+SH+NGRT vs CMS-CPCW/H+SH	NGRT	6	-0.002 [-0.002, -0.001] (p=1.0e+00)	-0.7 [-1.0, -0.4] (p=1.0e+00)	-0.004 [-0.011, +0.000]
CMS-CPCW/H+SH+NGRU vs CMS-CPCW/H+SH	NGRU	60	-0.000 [-0.001, -0.000] (p=3.9e-03)	-3.2 [-4.6, -2.1] (p=2.0e-03)	-0.004 [-0.006, -0.003]
CMS-CPCW/H+SH+NGR_ vs CMS-CPCW/H+SH	NGR_	60	-0.001 [-0.001, -0.000] (p=1.6e-02)	-8.2 [-15.8, -2.5] (p=3.9e-03)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+NSR vs CMS-CPCW/H+SH	NSR_office	60	-0.001 [-0.002, -0.000] (p=2.0e-03)	-0.6 [-0.8, -0.4] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+NSR vs CMS-CPCW/H+SH	NSR_old town	60	-0.009 [-0.012, -0.007] (p=2.0e-03)	-2.2 [-4.1, -0.3] (p=7.5e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+NSR vs CMS-CPCW/H+SH	NSR_countryside	60	-0.007 [-0.008, -0.006] (p=2.0e-03)	-0.4 [-2.9, +1.4] (p=7.5e-01)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+NGR vs CMS-CPC/H+SH+G	NGR_area	60	-0.001 [-0.001, -0.000] (p=3.9e-03)	-10.3 [-12.4, -8.4] (p=2.0e-03)	-0.003 [-0.004, -0.002]
CMS-CPC/H+SH+G+NGRT vs CMS-CPC/H+SH+G	NGRT	6	-0.002 [-0.003, -0.001] (p=1.0e+00)	-1.9 [-4.2, -0.3] (p=1.0e+00)	-0.006 [-0.018, +0.000]
CMS-CPC/H+SH+G+NGRU vs CMS-CPC/H+SH+G	NGRU	60	-0.000 [-0.001, -0.000] (p=1.6e-02)	-3.6 [-5.0, -2.4] (p=2.0e-03)	-0.008 [-0.010, -0.006]
CMS-CPC/H+SH+G+NGR_ vs CMS-CPC/H+SH+G	NGR_	60	-0.000 [-0.001, -0.000] (p=1.2e-01)	-9.0 [-15.4, -3.9] (p=2.0e-03)	-0.004 [-0.004, -0.002]
CMS-CPC/H+SH+G+NSR vs CMS-CPC/H+SH+G	NSR_office	60	-0.001 [-0.002, -0.000] (p=3.9e-03)	-0.8 [-1.2, -0.5] (p=2.1e-02)	-0.001 [-0.002, -0.001]
CMS-CPC/H+SH+G+NSR vs CMS-CPC/H+SH+G	NSR_old town	60	-0.009 [-0.012, -0.007] (p=2.0e-03)	-4.6 [-8.7, -1.1] (p=3.4e-01)	-0.002 [-0.003, -0.001]
CMS-CPC/H+SH+G+NSR vs CMS-CPC/H+SH+G	NSR_countryside	60	-0.007 [-0.009, -0.006] (p=2.0e-03)	-2.6 [-11.3, +2.2] (p=1.1e-01)	-0.002 [-0.003, -0.001]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR_area	60	-0.000 [-0.001, -0.000] (p=2.0e-03)	+0.5 [-2.3, +3.4] (p=1.0e+00)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+NGRT vs CMS-CPCW/H+SH	NGRT	6	-0.001 [-0.002, -0.001] (p=1.0e+00)	-0.9 [-1.4, -0.4] (p=1.0e+00)	-0.004 [-0.011, +0.000]
CMS-CPCW/H+SH+NGRU vs CMS-CPCW/H+SH	NGRU	60	-0.001 [-0.001, -0.000] (p=3.9e-03)	-2.3 [-3.1, -1.6] (p=2.0e-03)	-0.004 [-0.005, -0.002]
CMS-CPCW/H+SH+NGR_ vs CMS-CPCW/H+SH	NGR_	60	-0.000 [-0.001, -0.000] (p=1.6e-02)	-7.6 [-15.3, -2.1] (p=3.9e-03)	-0.002 [-0.002, -0.001]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR office	60	-0.001 [-0.001, -0.000] (p=2.0e-03)	-0.5 [-0.8, -0.3] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR old town	60	-0.006 [-0.007, -0.004] (p=2.0e-03)	-1.7 [-3.8, +0.0] (p=7.5e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR countryside	60	-0.005 [-0.006, -0.005] (p=2.0e-03)	-0.0 [-2.6, +1.6] (p=3.4e-01)	-0.001 [-0.001, -0.000]

f2

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+BGRe vs CPC-SC-T0.5+SH	BGRe areas	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	+0.0 [-1.0, +1.0] (p=7.5e-01)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGRT vs CPC-SC-T0.5+SH	NGRT	6	+0.022 [+0.009, +0.038] (p=1.0e+00)	-0.8 [-2.1, +0.2] (p=1.0e+00)	+0.001 [-0.002, +0.003]
CMS-CPCW/H+SH+NGRu vs CPC-SC-T0.5+SH	NGRu	60	+0.037 [+0.028, +0.047] (p=2.0e-03)	-0.8 [-1.8, +0.1] (p=7.5e-01)	-0.009 [-0.019, +0.000]
CMS-CPCW/H+SH+NGRd vs CPC-SC-T0.5+SH	NGRd land	60	+0.059 [+0.051, +0.066] (p=2.0e-03)	-4.0 [-6.7, -1.6] (p=2.1e-02)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.8 [-2.2, -1.4] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR old town	60	+0.059 [+0.049, +0.070] (p=2.0e-03)	-3.2 [-4.1, -2.3] (p=2.0e-03)	+0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR countryside	60	+0.135 [+0.112, +0.160] (p=2.0e-03)	-15.0 [-17.2, -12.9] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPC/H+SH+G+BGRe vs CPC-SC-T0.5+SH	BGRe areas	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	-1.3 [-2.4, -0.5] (p=2.1e-02)	+0.001 [-0.000, +0.001]
CMS-CPC/H+SH+G+NGRT vs CPC-SC-T0.5+SH	NGRT	6	+0.025 [-0.000, +0.053] (p=1.0e+00)	-1.0 [-2.2, -0.1] (p=1.0e+00)	+0.001 [-0.001, +0.004]
CMS-CPC/H+SH+G+NGRu vs CPC-SC-T0.5+SH	NGRu	60	+0.033 [+0.023, +0.044] (p=2.0e-03)	-0.9 [-1.9, -0.1] (p=1.0e+00)	-0.008 [-0.018, +0.001]
CMS-CPC/H+SH+G+NGRd vs CPC-SC-T0.5+SH	NGRd land	60	+0.053 [+0.044, +0.059] (p=2.0e-03)	-4.4 [-7.2, -2.1] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+MSR vs CPC-SC-T0.5+SH	MSR office	60	+0.062 [+0.053, +0.071] (p=2.0e-03)	-1.9 [-2.4, -1.6] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+G+MSR vs CPC-SC-T0.5+SH	MSR old town	60	+0.062 [+0.052, +0.074] (p=2.0e-03)	-4.0 [-5.0, -2.9] (p=2.0e-03)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+G+MSR vs CPC-SC-T0.5+SH	MSR countryside	60	+0.158 [+0.136, +0.182] (p=2.0e-03)	-18.5 [-20.5, -16.5] (p=2.0e-03)	+0.000 [-0.000, +0.001]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR Boreas	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	+1.9 [+0.5, +3.4] (p=1.1e-01)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR LT	6	+0.022 [+0.008, +0.038] (p=1.0e+00)	-0.2 [-1.9, +1.7] (p=1.0e+00)	+0.001 [-0.002, +0.003]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR RU	60	+0.037 [+0.028, +0.047] (p=2.0e-03)	-0.5 [-1.6, +0.5] (p=7.5e-01)	-0.008 [-0.019, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR Woodland	60	+0.059 [+0.051, +0.066] (p=2.0e-03)	-3.7 [-6.6, -1.1] (p=2.1e-02)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.7 [-2.2, -1.3] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR old town	60	+0.064 [+0.055, +0.075] (p=2.0e-03)	-2.8 [-4.1, -1.3] (p=2.1e-02)	+0.000 [-0.001, +0.001]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR countryside	60	+0.138 [+0.116, +0.163] (p=2.0e-03)	-14.8 [-17.3, -12.4] (p=2.0e-03)	+0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GNGR vs CPC-SC-T0.5+SH	GNGR Boreas	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	+0.0 [-1.0, +1.0] (p=7.5e-01)	+0.001 [+0.000, +0.002]
CMS-CPC/H+SH+GNGR vs CPC-SC-T0.5+SH	GNGR LT	6	+0.024 [-0.000, +0.053] (p=1.0e+00)	-0.4 [-2.1, +1.2] (p=1.0e+00)	+0.002 [-0.001, +0.005]
CMS-CPC/H+SH+GNGR vs CPC-SC-T0.5+SH	GNGR RU	60	+0.033 [+0.023, +0.044] (p=2.0e-03)	-0.7 [-1.7, +0.3] (p=7.5e-01)	-0.008 [-0.018, +0.001]
CMS-CPC/H+SH+GNGR vs CPC-SC-T0.5+SH	GNGR Woodland	60	+0.053 [+0.045, +0.059] (p=2.0e-03)	-4.5 [-7.3, -2.1] (p=2.0e-03)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR office	60	+0.062 [+0.053, +0.071] (p=2.0e-03)	-1.9 [-2.4, -1.5] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR old town	60	+0.068 [+0.058, +0.080] (p=2.0e-03)	-4.2 [-5.3, -3.1] (p=2.0e-03)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR countryside	60	+0.162 [+0.140, +0.186] (p=2.0e-03)	-19.0 [-21.0, -17.1] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR Boreas	60	-0.000 [-0.001, -0.000] (p=2.0e-03)	-2.7 [-3.9, -1.7] (p=2.0e-03)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR LT	6	-0.000 [-0.001, -0.000] (p=1.0e+00)	-1.1 [-1.9, -0.5] (p=1.0e+00)	-0.004 [-0.011, +0.000]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR RU	60	-0.000 [-0.000, -0.000] (p=7.8e-03)	-1.3 [-1.9, -0.8] (p=2.0e-03)	-0.004 [-0.006, -0.003]
CMS-CPCW/H+SH+NGR vs CMS-CPCW/H+SH	NGR Woodland	60	-0.001 [-0.002, -0.000] (p=6.2e-02)	-4.8 [-9.6, -1.3] (p=3.9e-03)	-0.002 [-0.002, -0.001]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR office	60	-0.001 [-0.002, -0.000] (p=2.0e-03)	-0.4 [-0.5, -0.2] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR old town	60	-0.012 [-0.014, -0.009] (p=2.0e-03)	-1.4 [-3.0, +0.1] (p=7.5e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR countryside	60	-0.009 [-0.011, -0.008] (p=2.0e-03)	+0.5 [-0.7, +1.8] (p=3.4e-01)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+G+BGR vs CMS-CPC/H+SH+G	BGR areas	60	-0.000 [-0.001, -0.000] (p=3.9e-03)	-6.0 [-7.2, -5.0] (p=2.0e-03)	-0.003 [-0.004, -0.002]
CMS-CPC/H+SH+G+NGLR vs CMS-CPC/H+SH+G	NGLR	6	-0.001 [-0.002, -0.000] (p=1.0e+00)	-1.7 [-3.6, -0.4] (p=1.0e+00)	-0.006 [-0.018, +0.000]
CMS-CPC/H+SH+G+CMLR vs CMS-CPC/H+SH+G	CMLR	60	-0.000 [-0.000, +0.000] (p=2.5e-01)	-1.5 [-2.2, -1.0] (p=2.0e-03)	-0.008 [-0.010, -0.005]
CMS-CPC/H+SH+G+NGRL vs CMS-CPC/H+SH+G	NGRL	60	-0.001 [-0.002, -0.000] (p=1.2e-01)	-3.6 [-6.0, -1.5] (p=2.0e-03)	-0.004 [-0.004, -0.002]
CMS-CPC/H+SH+G+MSG vs CMS-CPC/H+SH+G	MSG office	60	-0.001 [-0.002, -0.000] (p=3.9e-03)	-0.5 [-0.7, -0.2] (p=2.1e-02)	-0.001 [-0.002, -0.001]
CMS-CPC/H+SH+G+MSG vs CMS-CPC/H+SH+G	MSG old town	60	-0.012 [-0.014, -0.009] (p=2.0e-03)	-2.4 [-4.6, -0.4] (p=7.5e-01)	-0.002 [-0.003, -0.001]
CMS-CPC/H+SH+G+MSG vs CMS-CPC/H+SH+G	MSG countryside	60	-0.010 [-0.012, -0.009] (p=2.0e-03)	-1.1 [-6.9, +3.4] (p=1.1e-01)	-0.002 [-0.003, -0.001]
CMS-CPCW/H+SH+BGR vs CMS-CPCW/H+SH	BGR areas	60	-0.000 [-0.001, -0.000] (p=2.0e-03)	-0.9 [-2.3, +0.6] (p=3.4e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+NGLR vs CMS-CPCW/H+SH	NGLR	6	-0.001 [-0.001, -0.000] (p=1.0e+00)	-0.5 [-0.8, -0.2] (p=1.0e+00)	-0.004 [-0.011, +0.000]
CMS-CPCW/H+SH+CMLR vs CMS-CPCW/H+SH	CMLR	60	-0.000 [-0.001, -0.000] (p=3.9e-03)	-1.0 [-1.4, -0.7] (p=2.0e-03)	-0.004 [-0.005, -0.002]
CMS-CPCW/H+SH+NGRL vs CMS-CPCW/H+SH	NGRL	60	-0.001 [-0.001, -0.000] (p=3.1e-02)	-4.5 [-9.2, -1.2] (p=3.9e-03)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR office	60	-0.001 [-0.001, -0.000] (p=2.0e-03)	-0.3 [-0.4, -0.2] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR old town	60	-0.006 [-0.008, -0.004] (p=2.0e-03)	-0.9 [-2.2, +0.2] (p=7.5e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR countryside	60	-0.006 [-0.007, -0.005] (p=2.0e-03)	+0.7 [-0.4, +1.6] (p=3.4e-01)	-0.001 [-0.001, -0.000]

f3

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR_area	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	-18.0 [-26.0, -11.2] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGRT vs CPC-SC-T0.5+SH	NGRT	6	+0.020 [+0.007, +0.036] (p=1.0e+00)	-2.8 [-9.3, +2.7] (p=1.0e+00)	+0.002 [-0.002, +0.008]
CMS-CPCW/H+SH+NGRU vs CPC-SC-T0.5+SH	NGRU	60	+0.035 [+0.025, +0.045] (p=2.0e-03)	-1.8 [-4.3, +0.4] (p=7.5e-01)	-0.005 [-0.015, +0.003]
CMS-CPCW/H+SH+NGR_ vs CPC-SC-T0.5+SH	NGR_	60	+0.057 [+0.048, +0.064] (p=2.0e-03)	-83.4 [-163.8, -7.1] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR_office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.8 [-2.2, -1.4] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR_old town	60	+0.060 [+0.051, +0.071] (p=2.0e-03)	-50.8 [-58.1, -44.1] (p=2.0e-03)	+0.001 [-0.000, +0.002]
CMS-CPCW/H+SH+NSR vs CPC-SC-T0.5+SH	NSR_countryside	60	+0.133 [+0.111, +0.159] (p=2.0e-03)	-15.8 [-18.4, -13.5] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPC/H+SH+GNGR vs CPC-SC-T0.5+SH	GNGR_area	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	-18.3 [-26.8, -10.5] (p=2.1e-02)	+0.001 [+0.000, +0.002]
CMS-CPC/H+SH+GNGRT vs CPC-SC-T0.5+SH	GNGRT	6	+0.022 [-0.001, +0.050] (p=1.0e+00)	-4.9 [-14.4, +1.8] (p=1.0e+00)	+0.005 [-0.001, +0.015]
CMS-CPC/H+SH+GNGRU vs CPC-SC-T0.5+SH	GNGRU	60	+0.030 [+0.021, +0.041] (p=2.0e-03)	-1.9 [-4.0, -0.1] (p=3.4e-01)	-0.004 [-0.013, +0.005]
CMS-CPC/H+SH+GNGR_ vs CPC-SC-T0.5+SH	GNGR_	60	+0.048 [+0.039, +0.055] (p=2.0e-03)	-81.4 [-159.9, -7.5] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR_office	60	+0.061 [+0.053, +0.071] (p=2.0e-03)	-1.8 [-2.2, -1.4] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR_old town	60	+0.063 [+0.053, +0.075] (p=2.0e-03)	-51.8 [-59.0, -45.3] (p=2.0e-03)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+GNSR vs CPC-SC-T0.5+SH	GNSR_countryside	60	+0.156 [+0.134, +0.180] (p=2.0e-03)	-18.6 [-21.0, -16.1] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+NGR vs CPC-SC-T0.5+SH	NGR_area	60	+0.099 [+0.079, +0.116] (p=2.0e-03)	-18.5 [-26.2, -12.2] (p=2.0e-03)	+0.001 [-0.000, +0.001]
CMS-CPCW/H+SH+NGRT vs CPC-SC-T0.5+SH	NGRT	6	+0.021 [+0.008, +0.038] (p=1.0e+00)	-3.4 [-9.3, +0.9] (p=1.0e+00)	+0.002 [-0.002, +0.008]
CMS-CPCW/H+SH+NGRU vs CPC-SC-T0.5+SH	NGRU	60	+0.036 [+0.027, +0.046] (p=2.0e-03)	-2.3 [-5.1, +0.2] (p=7.5e-01)	-0.005 [-0.015, +0.003]
CMS-CPCW/H+SH+NGR_ vs CPC-SC-T0.5+SH	NGR_	60	+0.058 [+0.050, +0.066] (p=2.0e-03)	-83.7 [-164.6, -7.0] (p=2.0e-03)	-0.000 [-0.001, +0.000]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR office	60	+0.059 [+0.050, +0.069] (p=2.0e-03)	-1.7 [-2.2, -1.3] (p=2.0e-03)	+0.000 [+0.000, +0.000]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR old town	60	+0.064 [+0.055, +0.075] (p=2.0e-03)	-50.4 [-58.1, -43.3] (p=2.0e-03)	+0.001 [-0.000, +0.002]
CMS-CPCW/H+SH+MSR vs CPC-SC-T0.5+SH	MSR countryside	60	+0.137 [+0.115, +0.162] (p=2.0e-03)	-16.2 [-19.1, -13.5] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU boreas	60	+0.109 [+0.089, +0.125] (p=2.0e-03)	-19.7 [-27.3, -13.2] (p=2.0e-03)	+0.001 [+0.000, +0.002]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU NCLT	6	+0.023 [-0.001, +0.051] (p=1.0e+00)	-5.5 [-14.4, +0.5] (p=1.0e+00)	+0.005 [-0.001, +0.017]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU CMU	60	+0.032 [+0.022, +0.043] (p=2.0e-03)	-2.6 [-5.2, -0.4] (p=3.4e-01)	-0.004 [-0.013, +0.005]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU wildland	60	+0.051 [+0.043, +0.058] (p=2.0e-03)	-83.1 [-163.1, -7.5] (p=2.0e-03)	-0.000 [-0.001, +0.000]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU office	60	+0.062 [+0.053, +0.071] (p=2.0e-03)	-1.7 [-2.2, -1.3] (p=2.0e-03)	-0.000 [-0.000, +0.000]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU old town	60	+0.067 [+0.057, +0.079] (p=2.0e-03)	-52.2 [-59.5, -45.7] (p=2.0e-03)	-0.001 [-0.002, +0.001]
CMS-CPC/H+SH+GRU vs CPC-SC-T0.5+SH	GRU countryside	60	+0.160 [+0.138, +0.184] (p=2.0e-03)	-19.6 [-22.0, -17.1] (p=2.0e-03)	+0.000 [-0.000, +0.001]
CMS-CPCW/H+SH+GRU vs CMS-CPCW/H+SH	GRU boreas	60	-0.000 [-0.001, -0.000] (p=2.0e-03)	+5.8 [+2.5, +9.1] (p=1.1e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+GRU vs CMS-CPCW/H+SH	GRU NCLT	6	-0.003 [-0.007, -0.000] (p=1.0e+00)	+0.2 [-1.2, +2.4] (p=1.0e+00)	-0.002 [-0.007, +0.000]
CMS-CPCW/H+SH+GRU vs CMS-CPCW/H+SH	GRU CMU	60	-0.003 [-0.005, -0.001] (p=2.0e-03)	+0.9 [-0.1, +2.2] (p=3.4e-01)	-0.001 [-0.001, -0.000]
CMS-CPCW/H+SH+GRU vs CMS-CPCW/H+SH	GRU wildland	60	-0.003 [-0.006, -0.001] (p=3.9e-03)	+0.2 [-3.5, +3.8] (p=7.5e-01)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR office	60	-0.001 [-0.002, -0.001] (p=2.0e-03)	-0.2 [-0.2, -0.1] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR old town	60	-0.010 [-0.012, -0.008] (p=2.0e-03)	-0.4 [-1.9, +0.9] (p=7.5e-01)	-0.001 [-0.002, -0.000]
CMS-CPCW/H+SH+MSR vs CMS-CPCW/H+SH	MSR countryside	60	-0.011 [-0.012, -0.009] (p=2.0e-03)	+1.5 [+0.2, +2.5] (p=1.1e-01)	-0.001 [-0.001, -0.000]
CMS-CPC/H+SH+GRU vs CMS-CPC/H+SH+G	GRU boreas	60	-0.000 [-0.001, -0.000] (p=3.9e-03)	+3.0 [-1.4, +7.8] (p=7.5e-01)	-0.003 [-0.004, -0.002]

A vs B	dataset	runs	Δ recall [95 % CI] (sign p)	Δ error m [95 % CI] (sign p)	Δ segFAR [95 % CI]
CMS-CPC/H+SH+G+NCLR vs CMS-CPC/H+SH+G	NCLR	6	-0.004 [-0.009, -0.001] (p=1.0e+00)	+0.5 [-2.6, +4.8] (p=1.0e+00)	-0.002 [-0.007, +0.000]
CMS-CPC/H+SH+G+CNCLR vs CMS-CPC/H+SH+G	CNCLR	60	-0.003 [-0.006, -0.001] (p=2.0e-03)	+0.9 [-0.0, +2.3] (p=7.5e-01)	-0.003 [-0.005, -0.002]
CMS-CPC/H+SH+G+NCL vs CMS-CPC/H+SH+G	NCL	60	-0.006 [-0.012, -0.001] (p=1.6e-02)	+3.6 [-0.9, +9.4] (p=7.5e-01)	-0.003 [-0.004, -0.002]
CMS-CPC/H+SH+G+4SCR vs CMS-CPC/H+SH+G	4SCR	60	-0.001 [-0.002, -0.001] (p=2.0e-03)	-0.2 [-0.3, -0.1] (p=1.1e-01)	-0.001 [-0.002, -0.001]
CMS-CPC/H+SH+G+4SCR vs CMS-CPC/H+SH+G	4SCR	60	-0.011 [-0.013, -0.010] (p=2.0e-03)	-0.3 [-1.6, +0.9] (p=7.5e-01)	-0.002 [-0.003, -0.001]
CMS-CPC/H+SH+G+4SCR vs CMS-CPC/H+SH+G	4SCR	60	-0.012 [-0.015, -0.010] (p=2.0e-03)	+0.4 [-4.8, +4.4] (p=1.1e-01)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+NCR vs CMS-CPCW/H+SH	NCR	60	-0.000 [-0.000, -0.000] (p=2.0e-03)	+5.4 [+2.6, +8.2] (p=1.1e-01)	-0.001 [-0.002, -0.001]
CMS-CPCW/H+SH+NCLR vs CMS-CPCW/H+SH	NCLR	6	-0.001 [-0.003, -0.000] (p=1.0e+00)	-0.4 [-1.1, +0.6] (p=1.0e+00)	-0.002 [-0.007, +0.000]
CMS-CPCW/H+SH+NCRU vs CMS-CPCW/H+SH	NCRU	60	-0.001 [-0.002, -0.001] (p=2.0e-03)	+0.4 [-0.2, +1.1] (p=1.0e+00)	-0.001 [-0.001, -0.000]
CMS-CPCW/H+SH+NCL vs CMS-CPCW/H+SH	NCL	60	-0.002 [-0.003, -0.001] (p=3.9e-03)	-0.1 [-3.4, +3.1] (p=7.5e-01)	-0.002 [-0.002, -0.001]
CMS-CPCW/H+SH+4SCR vs CMS-CPCW/H+SH	4SCR	60	-0.001 [-0.001, -0.001] (p=2.0e-03)	-0.1 [-0.2, -0.1] (p=2.1e-02)	-0.001 [-0.001, -0.001]
CMS-CPCW/H+SH+4SCR vs CMS-CPCW/H+SH	4SCR	60	-0.006 [-0.008, -0.005] (p=2.0e-03)	+0.0 [-0.7, +0.7] (p=7.5e-01)	-0.001 [-0.002, -0.000]
CMS-CPCW/H+SH+4SCR vs CMS-CPCW/H+SH	4SCR	60	-0.007 [-0.008, -0.006] (p=2.0e-03)	+1.1 [+0.5, +1.8] (p=1.1e-01)	-0.001 [-0.001, -0.000]

Bảng 7. Off-map tự nhiên (4Seasons chéo cảnh, bản đồ A ← truy vấn B): recall cảnh A / segFAR TB / tệ nhất / sai số cảnh A (m)

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
Xu'21 (nhãn)	0.651 / 0.024 / 0.044 / 9.8	0.738 / 0.033 / 0.058 / 9.0	0.671 / 0.024 / 0.073 / 21.9	0.646 / 0.047 / 0.112 / 22.9	0.390 / 0.039 / 0.140 / 55.0	0.391 / 0.057 / 0.110 / 51.4
Claxton'24 (nhãn)	0.588 / 0.002 / 0.009 / 9.5	0.673 / 0.000 / 0.002 / 9.1	0.731 / 0.073 / 0.230 / 62.9	0.771 / 0.015 / 0.080 / 81.4	0.653 / 0.020 / 0.095 / 35.4	0.649 / 0.009 / 0.026 / 36.2
TAO'26 (nhãn)	0.201 / 0.000 / 0.000 / 31.1	0.201 / 0.000 / 0.000 / 31.1	0.809 / 0.297 / 0.999 / 72.8	0.809 / 0.166 / 0.994 / 72.8	0.726 / 0.001 / 0.004 / 41.7	0.726 / 0.000 / 0.000 / 41.7
CPC-SC+CS (vòng 3)	0.761 / 0.000 / 0.000 / 8.1	0.761 / 0.000 / 0.000 / 8.1	0.664 / 0.000 / 0.002 / 22.2	0.664 / 0.000 / 0.001 / 22.2	0.280 / 0.000 / 0.000 / 49.5	0.280 / 0.000 / 0.000 / 49.5
CPC-SC-T½+SH (vòng 5)	0.785 / 0.000 / 0.000 / 7.9	0.785 / 0.000 / 0.000 / 7.9	0.688 / 0.001 / 0.007 / 21.6	0.688 / 0.000 / 0.000 / 21.6	0.305 / 0.000 / 0.000 / 48.6	0.305 / 0.000 / 0.000 / 48.6

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CPC-SC-T½+SH+NGRr	–	–	–	–	–	–
CPC-SC-T½+SH+NGR	0.785 / 0.000 / 0.000 / 7.9	0.785 / 0.000 / 0.000 / 7.9	0.684 / 0.001 / 0.004 / 21.9	0.684 / 0.000 / 0.000 / 21.9	0.301 / 0.000 / 0.000 / 49.2	0.301 / 0.000 / 0.000 / 49.2
CMS-CPCW/H+SH [phiên (chuyển đạo)]	0.845 / 0.014 / 0.029 / 6.3	0.845 / 0.007 / 0.013 / 6.3	0.750 / 0.012 / 0.028 / 18.2	0.750 / 0.006 / 0.009 / 18.2	0.488 / 0.009 / 0.013 / 28.3	0.488 / 0.014 / 0.024 / 28.3
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)]	0.845 / 0.005 / 0.011 / 6.3	0.845 / 0.002 / 0.006 / 6.3	0.748 / 0.006 / 0.016 / 18.6	0.748 / 0.002 / 0.004 / 18.6	0.481 / 0.003 / 0.006 / 30.7	0.481 / 0.006 / 0.013 / 30.7
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)]	0.845 / 0.006 / 0.015 / 6.3	0.845 / 0.003 / 0.007 / 6.3	0.748 / 0.007 / 0.019 / 18.6	0.748 / 0.003 / 0.005 / 18.6	0.484 / 0.004 / 0.008 / 29.8	0.484 / 0.007 / 0.017 / 29.8
CMS-CPCW/H+SH [phiên khác]	0.832 / 0.004 / 0.007 / 6.6	0.838 / 0.003 / 0.010 / 6.5	0.765 / 0.001 / 0.006 / 18.1	0.768 / 0.001 / 0.004 / 18.5	0.540 / 0.004 / 0.014 / 31.0	0.524 / 0.007 / 0.021 / 31.9
CMS-CPCW/H+SH+NGRr [phiên khác]	0.831 / 0.001 / 0.004 / 6.6	0.838 / 0.001 / 0.004 / 6.5	0.764 / 0.001 / 0.003 / 18.2	0.767 / 0.001 / 0.002 / 18.6	0.534 / 0.001 / 0.003 / 33.4	0.519 / 0.003 / 0.009 / 35.2
CMS-CPCW/H+SH+NGR [phiên khác]	0.831 / 0.002 / 0.004 / 6.6	0.838 / 0.001 / 0.005 / 6.5	0.764 / 0.001 / 0.004 / 18.2	0.767 / 0.001 / 0.002 / 18.6	0.537 / 0.001 / 0.004 / 32.3	0.521 / 0.004 / 0.011 / 33.4
CMS-CPCW/H+SH [online nhân quả]	0.820 / 0.002 / 0.004 / 6.7	0.830 / 0.002 / 0.009 / 6.6	0.757 / 0.001 / 0.006 / 18.7	0.761 / 0.001 / 0.003 / 18.6	0.561 / 0.003 / 0.008 / 30.6	0.543 / 0.005 / 0.017 / 31.9
CMS-CPCW/H+SH+NGRr [online nhân quả]	0.820 / 0.000 / 0.001 / 6.7	0.830 / 0.001 / 0.003 / 6.6	0.756 / 0.001 / 0.002 / 18.8	0.759 / 0.001 / 0.002 / 18.7	0.556 / 0.001 / 0.002 / 32.6	0.538 / 0.002 / 0.008 / 33.9
CMS-CPCW/H+SH+NGR [online nhân quả]	0.820 / 0.000 / 0.002 / 6.7	0.830 / 0.001 / 0.004 / 6.6	0.756 / 0.001 / 0.003 / 18.8	0.759 / 0.001 / 0.002 / 18.7	0.558 / 0.001 / 0.004 / 31.8	0.540 / 0.003 / 0.009 / 32.5
CMS-CPCW/H+SH [giao phiên ∧ phiên khác]	0.826 / 0.002 / 0.006 / 6.6	0.830 / 0.003 / 0.010 / 6.6	0.740 / 0.001 / 0.006 / 18.9	0.740 / 0.001 / 0.004 / 19.3	0.455 / 0.002 / 0.006 / 35.5	0.444 / 0.005 / 0.015 / 36.1
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác]	0.826 / 0.001 / 0.004 / 6.6	0.830 / 0.001 / 0.004 / 6.6	0.738 / 0.001 / 0.003 / 19.2	0.738 / 0.001 / 0.002 / 19.6	0.449 / 0.001 / 0.002 / 37.8	0.437 / 0.002 / 0.007 / 39.3
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.826 / 0.001 / 0.004 / 6.6	0.830 / 0.001 / 0.005 / 6.6	0.738 / 0.001 / 0.004 / 19.2	0.738 / 0.001 / 0.002 / 19.6	0.452 / 0.001 / 0.003 / 36.8	0.440 / 0.003 / 0.008 / 37.6
CMS-CPCW/H+SH [phiên (chuyển đạo)], α = 0.01	0.809 / 0.005 / 0.011 / 7.0	0.809 / 0.003 / 0.007 / 7.0	0.708 / 0.007 / 0.018 / 19.9	0.708 / 0.003 / 0.007 / 19.9	0.351 / 0.001 / 0.004 / 52.3	0.351 / 0.004 / 0.007 / 52.3
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)], α = 0.01	0.808 / 0.002 / 0.004 / 7.0	0.808 / 0.001 / 0.003 / 7.0	0.705 / 0.003 / 0.012 / 20.2	0.705 / 0.001 / 0.003 / 20.2	0.342 / 0.001 / 0.002 / 54.7	0.342 / 0.002 / 0.003 / 54.7

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)], α = 0.01	0.808 / 0.003 / 0.007 / 7.0	0.808 / 0.001 / 0.005 / 7.0	0.706 / 0.004 / 0.014 / 20.1	0.706 / 0.001 / 0.005 / 20.1	0.346 / 0.001 / 0.002 / 53.5	0.346 / 0.002 / 0.003 / 53.5
CMS-CPCW/H+SH [phiên khác], α = 0.01	0.812 / 0.001 / 0.002 / 7.0	0.819 / 0.001 / 0.004 / 6.8	0.732 / 0.001 / 0.004 / 20.2	0.746 / 0.000 / 0.001 / 20.0	0.469 / 0.001 / 0.006 / 44.8	0.459 / 0.002 / 0.009 / 46.1
CMS-CPCW/H+SH+NGRr [phiên khác], α = 0.01	0.812 / 0.000 / 0.001 / 7.1	0.819 / 0.001 / 0.002 / 6.8	0.730 / 0.000 / 0.001 / 20.3	0.744 / 0.000 / 0.000 / 20.1	0.463 / 0.000 / 0.002 / 45.7	0.454 / 0.001 / 0.004 / 47.0
CMS-CPCW/H+SH+NGR [phiên khác], α = 0.01	0.812 / 0.000 / 0.001 / 7.1	0.819 / 0.001 / 0.002 / 6.8	0.730 / 0.000 / 0.002 / 20.3	0.744 / 0.000 / 0.000 / 20.1	0.466 / 0.001 / 0.004 / 45.3	0.457 / 0.001 / 0.006 / 46.6
CMS-CPCW/H+SH [online nhân quả], α = 0.01	0.804 / 0.001 / 0.002 / 7.1	0.813 / 0.001 / 0.003 / 6.9	0.732 / 0.000 / 0.002 / 20.4	0.739 / 0.000 / 0.001 / 20.0	0.488 / 0.000 / 0.000 / 44.7	0.476 / 0.001 / 0.004 / 47.2
CMS-CPCW/H+SH+NGRr [online nhân quả], α = 0.01	0.803 / 0.000 / 0.001 / 7.2	0.813 / 0.000 / 0.000 / 6.9	0.731 / 0.000 / 0.001 / 20.8	0.737 / 0.000 / 0.000 / 20.2	0.482 / 0.000 / 0.000 / 45.5	0.470 / 0.000 / 0.002 / 48.2
CMS-CPCW/H+SH+NGR [online nhân quả], α = 0.01	0.803 / 0.000 / 0.001 / 7.2	0.813 / 0.000 / 0.001 / 6.9	0.731 / 0.000 / 0.001 / 20.8	0.737 / 0.000 / 0.000 / 20.2	0.485 / 0.000 / 0.000 / 45.1	0.474 / 0.000 / 0.003 / 47.7
CMS-CPCW/H+SH [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.001 / 0.002 / 7.2	0.803 / 0.001 / 0.001 / 7.1	0.700 / 0.001 / 0.003 / 21.2	0.702 / 0.000 / 0.001 / 21.1	0.316 / 0.000 / 0.000 / 62.4	0.314 / 0.001 / 0.004 / 62.9
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.000 / 0.001 / 7.2	0.803 / 0.000 / 0.001 / 7.1	0.699 / 0.000 / 0.001 / 21.4	0.700 / 0.000 / 0.000 / 21.3	0.308 / 0.000 / 0.000 / 63.5	0.306 / 0.000 / 0.002 / 64.1
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.000 / 0.001 / 7.2	0.803 / 0.000 / 0.001 / 7.1	0.699 / 0.000 / 0.002 / 21.4	0.700 / 0.000 / 0.000 / 21.2	0.312 / 0.000 / 0.000 / 62.9	0.310 / 0.000 / 0.003 / 63.5
CMS-CPC/H+SH+G [phiên (chuyển đạo)]	0.845 / 0.022 / 0.044 / 6.3	0.845 / 0.011 / 0.026 / 6.3	0.752 / 0.015 / 0.028 / 17.9	0.752 / 0.012 / 0.018 / 17.9	0.512 / 0.014 / 0.023 / 24.5	0.512 / 0.024 / 0.043 / 24.5
CMS-CPC/H+SH+G+NGRr [phiên (chuyển đạo)]	0.845 / 0.005 / 0.013 / 6.3	0.845 / 0.004 / 0.009 / 6.3	0.749 / 0.005 / 0.012 / 18.2	0.749 / 0.004 / 0.007 / 18.2	0.504 / 0.003 / 0.008 / 28.9	0.504 / 0.006 / 0.010 / 28.9
CMS-CPC/H+SH+G+NGR [phiên (chuyển đạo)]	0.845 / 0.007 / 0.016 / 6.3	0.845 / 0.005 / 0.013 / 6.3	0.749 / 0.006 / 0.015 / 18.2	0.749 / 0.005 / 0.008 / 18.2	0.506 / 0.003 / 0.008 / 28.1	0.506 / 0.007 / 0.013 / 28.1

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CMS-CPC/H+SH+G [phiên khác]	0.837 / 0.014 / 0.021 / 6.5	0.844 / 0.007 / 0.023 / 6.4	0.770 / 0.008 / 0.017 / 28.0	0.774 / 0.005 / 0.009 / 17.8	0.546 / 0.011 / 0.027 / 29.2	0.541 / 0.023 / 0.035 / 29.7
CMS-CPC/H+SH+G+NGRr [phiên khác]	0.837 / 0.004 / 0.010 / 6.5	0.843 / 0.002 / 0.008 / 6.4	0.769 / 0.002 / 0.006 / 17.6	0.772 / 0.002 / 0.003 / 17.8	0.539 / 0.002 / 0.005 / 31.7	0.535 / 0.004 / 0.007 / 32.3
CMS-CPC/H+SH+G+NGR [phiên khác]	0.837 / 0.006 / 0.013 / 6.5	0.844 / 0.003 / 0.010 / 6.4	0.769 / 0.002 / 0.007 / 17.6	0.772 / 0.002 / 0.004 / 17.8	0.541 / 0.003 / 0.007 / 30.8	0.537 / 0.006 / 0.010 / 31.3
CMS-CPC/H+SH+G [online nhân quả]	0.821 / 0.002 / 0.004 / 6.7	0.829 / 0.002 / 0.009 / 6.6	0.755 / 0.001 / 0.005 / 18.5	0.759 / 0.001 / 0.003 / 18.6	0.556 / 0.004 / 0.012 / 30.5	0.545 / 0.006 / 0.020 / 31.1
CMS-CPC/H+SH+G+NGRr [online nhân quả]	0.821 / 0.001 / 0.001 / 6.7	0.828 / 0.001 / 0.003 / 6.6	0.754 / 0.001 / 0.002 / 18.7	0.756 / 0.001 / 0.002 / 18.8	0.550 / 0.001 / 0.003 / 31.0	0.539 / 0.002 / 0.004 / 31.5
CMS-CPC/H+SH+G+NGR [online nhân quả]	0.821 / 0.001 / 0.002 / 6.7	0.828 / 0.001 / 0.004 / 6.6	0.754 / 0.001 / 0.002 / 18.7	0.756 / 0.001 / 0.002 / 18.8	0.552 / 0.001 / 0.004 / 30.7	0.541 / 0.002 / 0.007 / 31.2
CMS-CPC/H+SH+G [giao phiên ∧ phiên khác]	0.833 / 0.013 / 0.020 / 6.5	0.836 / 0.007 / 0.022 / 6.4	0.748 / 0.007 / 0.016 / 17.9	0.748 / 0.005 / 0.009 / 18.1	0.482 / 0.008 / 0.018 / 31.8	0.477 / 0.017 / 0.030 / 32.1
CMS-CPC/H+SH+G+NGRr [giao phiên ∧ phiên khác]	0.833 / 0.004 / 0.010 / 6.5	0.836 / 0.002 / 0.007 / 6.4	0.747 / 0.002 / 0.006 / 18.1	0.746 / 0.002 / 0.003 / 18.3	0.475 / 0.002 / 0.004 / 34.1	0.471 / 0.004 / 0.007 / 34.5
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.833 / 0.005 / 0.013 / 6.5	0.836 / 0.003 / 0.010 / 6.4	0.747 / 0.002 / 0.007 / 18.1	0.746 / 0.002 / 0.004 / 18.3	0.477 / 0.002 / 0.005 / 33.2	0.473 / 0.005 / 0.009 / 33.7

Gộp 36 run: recall cảnh A / segFAR TB / segFAR tệ nhất / tỷ lệ run segFAR > α / sai số cảnh A (m)

Luật	gộp
Xu'21 (nhãn)	0.581 / 0.037 / 0.140 / 0.28 / 28.3
Claxton'24 (nhãn)	0.678 / 0.020 / 0.230 / 0.14 / 39.1
TAO'26 (nhãn)	0.579 / 0.077 / 0.999 / 0.11 / 48.5
CPC-SC+CS (vòng 3)	0.568 / 0.000 / 0.002 / 0.00 / 26.6
CPC-SC-T½+SH (vòng 5)	0.593 / 0.000 / 0.007 / 0.00 / 26.0
CPC-SC-T½+SH+NGR	0.590 / 0.000 / 0.004 / 0.00 / 26.3
CMS-CPCW/H+SH [phiên (chuyển đạo)]	0.695 / 0.010 / 0.029 / 0.00 / 17.6
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)]	0.691 / 0.004 / 0.016 / 0.00 / 18.6
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)]	0.692 / 0.005 / 0.019 / 0.00 / 18.2
CMS-CPCW/H+SH [phiên khác]	0.711 / 0.003 / 0.021 / 0.00 / 18.8
CMS-CPCW/H+SH+NGRr [phiên khác]	0.709 / 0.001 / 0.009 / 0.00 / 19.7
CMS-CPCW/H+SH+NGR [phiên khác]	0.710 / 0.002 / 0.011 / 0.00 / 19.3
CMS-CPCW/H+SH [online nhân quả]	0.712 / 0.003 / 0.017 / 0.00 / 18.8
CMS-CPCW/H+SH+NGRr [online nhân quả]	0.710 / 0.001 / 0.008 / 0.00 / 19.6
CMS-CPCW/H+SH+NGR [online nhân quả]	0.710 / 0.001 / 0.009 / 0.00 / 19.2
CMS-CPCW/H+SH [giao phiên ∧ phiên khác]	0.673 / 0.002 / 0.015 / 0.00 / 20.5

Luật	gộp
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác]	0.670 / 0.001 / 0.007 / 0.00 / 21.5
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.671 / 0.001 / 0.008 / 0.00 / 21.1
CMS-CPCW/H+SH [phiên (chuyển đạo)], α = 0.01	0.622 / 0.004 / 0.018 / 0.06 / 26.4
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)], α = 0.01	0.619 / 0.002 / 0.012 / 0.03 / 27.3
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)], α = 0.01	0.620 / 0.002 / 0.014 / 0.03 / 26.9
CMS-CPCW/H+SH [phiên khác], α = 0.01	0.673 / 0.001 / 0.009 / 0.00 / 24.2
CMS-CPCW/H+SH+NGRr [phiên khác], α = 0.01	0.670 / 0.000 / 0.004 / 0.00 / 24.5
CMS-CPCW/H+SH+NGR [phiên khác], α = 0.01	0.671 / 0.001 / 0.006 / 0.00 / 24.4
CMS-CPCW/H+SH [online nhân quả], α = 0.01	0.675 / 0.001 / 0.004 / 0.00 / 24.4
CMS-CPCW/H+SH+NGRr [online nhân quả], α = 0.01	0.673 / 0.000 / 0.002 / 0.00 / 24.8
CMS-CPCW/H+SH+NGR [online nhân quả], α = 0.01	0.674 / 0.000 / 0.003 / 0.00 / 24.6
CMS-CPCW/H+SH [giao phiên ∧ phiên khác], α = 0.01	0.606 / 0.000 / 0.004 / 0.00 / 30.3
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác], α = 0.01	0.602 / 0.000 / 0.002 / 0.00 / 30.8
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác], α = 0.01	0.604 / 0.000 / 0.003 / 0.00 / 30.6
CMS-CPC/H+SH+G [phiên (chuyển đạo)]	0.703 / 0.016 / 0.044 / 0.00 / 16.2
CMS-CPC/H+SH+G+NGRr [phiên (chuyển đạo)]	0.699 / 0.004 / 0.013 / 0.00 / 17.8
CMS-CPC/H+SH+G+NGR [phiên (chuyển đạo)]	0.700 / 0.005 / 0.016 / 0.00 / 17.5
CMS-CPC/H+SH+G [phiên khác]	0.719 / 0.011 / 0.035 / 0.00 / 19.6
CMS-CPC/H+SH+G+NGRr [phiên khác]	0.716 / 0.003 / 0.010 / 0.00 / 18.7
CMS-CPC/H+SH+G+NGR [phiên khác]	0.717 / 0.003 / 0.013 / 0.00 / 18.4
CMS-CPC/H+SH+G [online nhân quả]	0.711 / 0.003 / 0.020 / 0.00 / 18.6
CMS-CPC/H+SH+G+NGRr [online nhân quả]	0.708 / 0.001 / 0.004 / 0.00 / 18.9
CMS-CPC/H+SH+G+NGR [online nhân quả]	0.709 / 0.001 / 0.007 / 0.00 / 18.8
CMS-CPC/H+SH+G [giao phiên ∧ phiên khác]	0.687 / 0.009 / 0.030 / 0.00 / 18.8
CMS-CPC/H+SH+G+NGRr [giao phiên ∧ phiên khác]	0.685 / 0.003 / 0.010 / 0.00 / 19.7
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.685 / 0.003 / 0.013 / 0.00 / 19.4

Bảng 8. Off-map tự nhiên, khởi tạo không nhãn (F3)

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
Xu'21 (nhãn)	0.651 / 0.024 / 0.044 / 9.8	0.738 / 0.033 / 0.058 / 8.9	0.671 / 0.024 / 0.073 / 72.9	0.646 / 0.047 / 0.112 / 79.1	0.390 / 0.039 / 0.140 / 55.6	0.391 / 0.057 / 0.110 / 51.4
Claxton'24 (nhãn)	0.588 / 0.002 / 0.009 / 9.5	0.673 / 0.000 / 0.002 / 9.0	0.731 / 0.073 / 0.230 / 66.9	0.771 / 0.015 / 0.080 / 87.3	0.653 / 0.020 / 0.095 / 35.5	0.649 / 0.009 / 0.026 / 36.3
TAO'26 (nhãn)	0.201 / 0.000 / 0.000 / 31.5	0.201 / 0.000 / 0.000 / 31.5	0.809 / 0.297 / 0.999 / 75.0	0.809 / 0.166 / 0.994 / 75.0	0.726 / 0.001 / 0.004 / 41.7	0.726 / 0.000 / 0.000 / 41.7
CPC-SC+CS (vòng 3)	0.761 / 0.000 / 0.000 / 8.0	0.761 / 0.000 / 0.000 / 8.0	0.664 / 0.000 / 0.002 / 97.9	0.664 / 0.000 / 0.001 / 97.9	0.280 / 0.000 / 0.000 / 50.4	0.280 / 0.000 / 0.000 / 50.4
CPC-SC-T½+SH (vòng 5)	0.785 / 0.000 / 0.000 / 7.7	0.785 / 0.000 / 0.000 / 7.7	0.688 / 0.001 / 0.007 / 96.7	0.688 / 0.000 / 0.000 / 96.7	0.305 / 0.000 / 0.000 / 50.2	0.305 / 0.000 / 0.000 / 50.2
CPC-SC-T½+SH+NGRr	–	–	–	–	–	–
CPC-SC-T½+SH+NGR	0.785 / 0.000 / 0.000 / 7.7	0.785 / 0.000 / 0.000 / 7.7	0.684 / 0.001 / 0.007 / 97.1	0.684 / 0.000 / 0.000 / 97.1	0.301 / 0.000 / 0.000 / 51.3	0.301 / 0.000 / 0.000 / 51.3
CMS-CPCW/H+SH [phiên (chuyển đạo)]	0.845 / 0.014 / 0.029 / 6.3	0.845 / 0.007 / 0.013 / 6.3	0.750 / 0.012 / 0.028 / 56.4	0.750 / 0.006 / 0.009 / 56.4	0.488 / 0.009 / 0.013 / 28.3	0.488 / 0.014 / 0.024 / 28.3

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CMS-CPCW/ H+SH+NGRr [phiên (chuyển đạo)]	0.845 / 0.008 / 0.017 / 6.4	0.845 / 0.004 / 0.009 / 6.4	0.747 / 0.009 / 0.023 / 56.8	0.747 / 0.004 / 0.006 / 56.8	0.481 / 0.007 / 0.012 / 30.7	0.481 / 0.008 / 0.018 / 30.7
CMS-CPCW/ H+SH+NGR [phiên (chuyển đạo)]	0.845 / 0.009 / 0.019 / 6.3	0.845 / 0.005 / 0.010 / 6.3	0.748 / 0.009 / 0.024 / 56.7	0.748 / 0.004 / 0.007 / 56.7	0.484 / 0.007 / 0.012 / 29.8	0.484 / 0.009 / 0.020 / 29.8
CMS-CPCW/ H+SH [phiên khác]	0.832 / 0.004 / 0.007 / 6.6	0.838 / 0.003 / 0.010 / 6.5	0.765 / 0.001 / 0.006 / 55.1	0.768 / 0.001 / 0.004 / 55.5	0.540 / 0.004 / 0.014 / 31.0	0.524 / 0.007 / 0.021 / 31.9
CMS-CPCW/ H+SH+NGRr [phiên khác]	0.831 / 0.003 / 0.005 / 6.6	0.838 / 0.002 / 0.007 / 6.5	0.764 / 0.001 / 0.006 / 55.2	0.767 / 0.001 / 0.004 / 55.6	0.534 / 0.003 / 0.009 / 33.4	0.519 / 0.004 / 0.012 / 35.2
CMS-CPCW/ H+SH+NGR [phiên khác]	0.831 / 0.003 / 0.005 / 6.6	0.838 / 0.002 / 0.007 / 6.5	0.764 / 0.001 / 0.006 / 55.2	0.767 / 0.001 / 0.004 / 55.6	0.537 / 0.003 / 0.011 / 32.3	0.521 / 0.005 / 0.014 / 33.4
CMS-CPCW/ H+SH [online nhân quả]	0.820 / 0.002 / 0.004 / 6.7	0.830 / 0.002 / 0.009 / 6.6	0.757 / 0.001 / 0.006 / 55.8	0.761 / 0.001 / 0.003 / 55.4	0.561 / 0.003 / 0.008 / 30.6	0.543 / 0.005 / 0.017 / 31.9
CMS-CPCW/ H+SH+NGRr [online nhân quả]	0.820 / 0.002 / 0.004 / 6.7	0.830 / 0.002 / 0.007 / 6.6	0.756 / 0.001 / 0.005 / 55.9	0.759 / 0.001 / 0.003 / 55.5	0.556 / 0.002 / 0.006 / 32.5	0.538 / 0.004 / 0.010 / 33.9
CMS-CPCW/ H+SH+NGR [online nhân quả]	0.820 / 0.002 / 0.004 / 6.7	0.830 / 0.002 / 0.007 / 6.6	0.756 / 0.001 / 0.005 / 55.9	0.759 / 0.001 / 0.003 / 55.5	0.558 / 0.002 / 0.007 / 31.8	0.540 / 0.004 / 0.011 / 32.5
CMS-CPCW/ H+SH [giao phiên ∧ phiên khác]	0.826 / 0.002 / 0.006 / 6.6	0.830 / 0.003 / 0.010 / 6.6	0.740 / 0.001 / 0.006 / 57.6	0.740 / 0.001 / 0.004 / 57.9	0.455 / 0.002 / 0.006 / 35.5	0.444 / 0.005 / 0.015 / 36.1
CMS-CPCW/ H+SH+NGRr [giao phiên ∧ phiên khác]	0.826 / 0.002 / 0.005 / 6.6	0.830 / 0.002 / 0.007 / 6.6	0.738 / 0.001 / 0.006 / 57.9	0.738 / 0.001 / 0.004 / 58.2	0.449 / 0.002 / 0.005 / 37.8	0.437 / 0.004 / 0.010 / 39.3
CMS-CPCW/ H+SH+NGR [giao phiên ∧ phiên khác]	0.826 / 0.002 / 0.005 / 6.6	0.830 / 0.002 / 0.007 / 6.6	0.738 / 0.001 / 0.006 / 57.9	0.738 / 0.001 / 0.004 / 58.2	0.452 / 0.002 / 0.005 / 36.8	0.440 / 0.004 / 0.010 / 37.6
CMS-CPCW/ H+SH [phiên (chuyển đạo)], α = 0.01	0.809 / 0.005 / 0.011 / 7.0	0.809 / 0.003 / 0.007 / 7.0	0.708 / 0.007 / 0.018 / 61.9	0.708 / 0.003 / 0.007 / 61.9	0.351 / 0.001 / 0.004 / 52.2	0.351 / 0.004 / 0.007 / 52.2
CMS-CPCW/ H+SH+NGRr [phiên (chuyển đạo)], α = 0.01	0.808 / 0.004 / 0.007 / 7.0	0.808 / 0.002 / 0.006 / 7.0	0.705 / 0.007 / 0.017 / 62.2	0.705 / 0.002 / 0.005 / 62.2	0.342 / 0.001 / 0.004 / 54.7	0.342 / 0.004 / 0.006 / 54.7
CMS-CPCW/ H+SH+NGR [phiên (chuyển đạo)], α = 0.01	0.808 / 0.004 / 0.009 / 7.0	0.808 / 0.002 / 0.007 / 7.0	0.706 / 0.007 / 0.017 / 62.2	0.706 / 0.002 / 0.006 / 62.2	0.346 / 0.001 / 0.004 / 53.4	0.346 / 0.004 / 0.006 / 53.4

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CMS-CPCW/ H+SH [phiên khác], α = 0.01	0.812 / 0.001 / 0.002 / 7.0	0.819 / 0.001 / 0.004 / 6.8	0.732 / 0.001 / 0.004 / 78.6	0.746 / 0.000 / 0.001 / 69.1	0.469 / 0.001 / 0.006 / 44.8	0.459 / 0.002 / 0.009 / 46.1
CMS-CPCW/ H+SH+NGRr [phiên khác], α = 0.01	0.812 / 0.001 / 0.002 / 7.0	0.819 / 0.001 / 0.004 / 6.8	0.730 / 0.001 / 0.003 / 78.7	0.744 / 0.000 / 0.001 / 69.2	0.463 / 0.001 / 0.006 / 45.7	0.454 / 0.001 / 0.007 / 47.0
CMS-CPCW/ H+SH+NGR [phiên khác], α = 0.01	0.812 / 0.001 / 0.002 / 7.0	0.819 / 0.001 / 0.004 / 6.8	0.731 / 0.001 / 0.003 / 78.7	0.744 / 0.000 / 0.001 / 69.2	0.466 / 0.001 / 0.006 / 45.3	0.457 / 0.001 / 0.007 / 46.6
CMS-CPCW/ H+SH [online nhân quả], α = 0.01	0.804 / 0.001 / 0.002 / 7.1	0.813 / 0.001 / 0.003 / 6.9	0.732 / 0.000 / 0.002 / 96.0	0.739 / 0.000 / 0.001 / 96.7	0.488 / 0.000 / 0.000 / 44.7	0.476 / 0.001 / 0.004 / 47.2
CMS-CPCW/ H+SH+NGRr [online nhân quả], α = 0.01	0.803 / 0.001 / 0.002 / 7.2	0.813 / 0.001 / 0.003 / 6.9	0.731 / 0.000 / 0.002 / 96.1	0.737 / 0.000 / 0.001 / 96.9	0.482 / 0.000 / 0.000 / 45.5	0.470 / 0.001 / 0.004 / 48.1
CMS-CPCW/ H+SH+NGR [online nhân quả], α = 0.01	0.803 / 0.001 / 0.002 / 7.2	0.813 / 0.001 / 0.003 / 6.9	0.731 / 0.000 / 0.002 / 96.1	0.737 / 0.000 / 0.001 / 96.9	0.485 / 0.000 / 0.000 / 45.1	0.474 / 0.001 / 0.004 / 47.7
CMS-CPCW/ H+SH [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.001 / 0.002 / 7.2	0.803 / 0.001 / 0.001 / 7.0	0.700 / 0.001 / 0.003 / 80.9	0.702 / 0.000 / 0.001 / 99.3	0.316 / 0.000 / 0.000 / 62.3	0.314 / 0.001 / 0.004 / 62.9
CMS-CPCW/ H+SH+NGRr [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.001 / 0.002 / 7.2	0.803 / 0.000 / 0.001 / 7.1	0.699 / 0.001 / 0.003 / 81.1	0.700 / 0.000 / 0.001 / 99.5	0.308 / 0.000 / 0.000 / 63.5	0.306 / 0.001 / 0.004 / 64.0
CMS-CPCW/ H+SH+NGR [giao phiên ∧ phiên khác], α = 0.01	0.798 / 0.001 / 0.002 / 7.2	0.803 / 0.000 / 0.001 / 7.1	0.699 / 0.001 / 0.003 / 81.1	0.700 / 0.000 / 0.001 / 99.4	0.312 / 0.000 / 0.000 / 62.9	0.310 / 0.001 / 0.004 / 63.5
CMS-CPC/ H+SH+G [phiên (chuyển đạo)]	0.845 / 0.022 / 0.044 / 6.3	0.845 / 0.011 / 0.026 / 6.3	0.752 / 0.015 / 0.028 / 54.7	0.752 / 0.012 / 0.018 / 54.7	0.512 / 0.014 / 0.023 / 24.5	0.512 / 0.024 / 0.043 / 24.5
CMS-CPC/ H+SH+G+NGRr [phiên (chuyển đạo)]	0.845 / 0.010 / 0.021 / 6.3	0.845 / 0.005 / 0.012 / 6.3	0.749 / 0.011 / 0.023 / 55.1	0.749 / 0.007 / 0.010 / 55.1	0.504 / 0.009 / 0.015 / 28.9	0.504 / 0.010 / 0.016 / 28.9
CMS-CPC/ H+SH+G+NGR [phiên (chuyển đạo)]	0.845 / 0.011 / 0.024 / 6.3	0.845 / 0.006 / 0.014 / 6.3	0.749 / 0.011 / 0.025 / 55.1	0.749 / 0.007 / 0.011 / 55.1	0.506 / 0.009 / 0.015 / 28.1	0.506 / 0.011 / 0.018 / 28.1
CMS-CPC/ H+SH+G [phiên khác]	0.837 / 0.014 / 0.021 / 6.5	0.844 / 0.007 / 0.023 / 6.4	0.770 / 0.008 / 0.017 / 56.0	0.774 / 0.005 / 0.009 / 54.5	0.546 / 0.011 / 0.027 / 29.2	0.541 / 0.023 / 0.035 / 29.7

Luật	office ← old town	office ← countryside	old town ← office	old town ← countryside	countryside ← office	countryside ← old town
CMS-CPC/H+SH+G+NGRr [phiên khác]	0.837 / 0.008 / 0.015 / 6.5	0.843 / 0.004 / 0.012 / 6.4	0.769 / 0.006 / 0.012 / 57.3	0.772 / 0.003 / 0.006 / 54.5	0.539 / 0.006 / 0.015 / 31.7	0.535 / 0.008 / 0.013 / 32.3
CMS-CPC/H+SH+G+NGR [phiên khác]	0.837 / 0.009 / 0.016 / 6.5	0.844 / 0.004 / 0.013 / 6.4	0.769 / 0.006 / 0.012 / 57.3	0.772 / 0.003 / 0.007 / 54.5	0.541 / 0.007 / 0.017 / 30.8	0.537 / 0.009 / 0.013 / 31.3
CMS-CPC/H+SH+G [online nhân quả]	0.821 / 0.002 / 0.004 / 6.7	0.829 / 0.002 / 0.009 / 6.6	0.755 / 0.001 / 0.005 / 55.3	0.759 / 0.001 / 0.003 / 55.4	0.556 / 0.004 / 0.012 / 30.5	0.545 / 0.006 / 0.020 / 31.1
CMS-CPC/H+SH+G+NGRr [online nhân quả]	0.821 / 0.002 / 0.004 / 6.7	0.828 / 0.001 / 0.007 / 6.6	0.754 / 0.001 / 0.004 / 55.5	0.757 / 0.001 / 0.003 / 55.6	0.550 / 0.003 / 0.009 / 31.0	0.539 / 0.003 / 0.009 / 31.5
CMS-CPC/H+SH+G+NGR [online nhân quả]	0.821 / 0.002 / 0.004 / 6.7	0.828 / 0.002 / 0.007 / 6.6	0.754 / 0.001 / 0.004 / 55.5	0.757 / 0.001 / 0.003 / 55.6	0.552 / 0.003 / 0.010 / 30.7	0.541 / 0.004 / 0.011 / 31.2
CMS-CPC/H+SH+G [giao phiên ∧ phiên khác]	0.833 / 0.013 / 0.020 / 6.5	0.836 / 0.007 / 0.022 / 6.4	0.748 / 0.007 / 0.016 / 54.5	0.748 / 0.005 / 0.009 / 55.3	0.482 / 0.008 / 0.018 / 31.8	0.477 / 0.017 / 0.030 / 32.1
CMS-CPC/H+SH+G+NGRr [giao phiên ∧ phiên khác]	0.833 / 0.008 / 0.015 / 6.5	0.836 / 0.004 / 0.011 / 6.4	0.747 / 0.006 / 0.012 / 54.6	0.746 / 0.003 / 0.006 / 55.5	0.475 / 0.006 / 0.014 / 34.0	0.471 / 0.008 / 0.013 / 34.5
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.833 / 0.008 / 0.016 / 6.5	0.836 / 0.004 / 0.013 / 6.4	0.747 / 0.006 / 0.012 / 54.6	0.746 / 0.003 / 0.007 / 55.5	0.477 / 0.006 / 0.014 / 33.2	0.473 / 0.008 / 0.013 / 33.7

Gộp 36 run: recall cảnh A / segFAR TB / segFAR tệ nhất / tỷ lệ run segFAR > α / sai số cảnh A (m)

Luật	gộp
Xu'21 (nhãn)	0.581 / 0.037 / 0.140 / 0.28 / 46.3
Claxton'24 (nhãn)	0.678 / 0.020 / 0.230 / 0.14 / 40.8
TAO'26 (nhãn)	0.579 / 0.077 / 0.999 / 0.11 / 49.4
CPC-SC+CS (vòng 3)	0.568 / 0.000 / 0.002 / 0.00 / 52.1
CPC-SC-T½+SH (vòng 5)	0.593 / 0.000 / 0.007 / 0.00 / 51.5
CPC-SC-T½+SH+NGR	0.590 / 0.000 / 0.007 / 0.00 / 52.0
CMS-CPCW/H+SH [phiên (chuyển đạo)]	0.695 / 0.010 / 0.029 / 0.00 / 30.4
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)]	0.691 / 0.007 / 0.023 / 0.00 / 31.3
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)]	0.692 / 0.007 / 0.024 / 0.00 / 31.0
CMS-CPCW/H+SH [phiên khác]	0.711 / 0.003 / 0.021 / 0.00 / 31.1
CMS-CPCW/H+SH+NGRr [phiên khác]	0.709 / 0.002 / 0.012 / 0.00 / 32.1
CMS-CPCW/H+SH+NGR [phiên khác]	0.710 / 0.003 / 0.014 / 0.00 / 31.6
CMS-CPCW/H+SH [online nhân quả]	0.712 / 0.003 / 0.017 / 0.00 / 31.2
CMS-CPCW/H+SH+NGRr [online nhân quả]	0.710 / 0.002 / 0.010 / 0.00 / 31.9
CMS-CPCW/H+SH+NGR [online nhân quả]	0.710 / 0.002 / 0.011 / 0.00 / 31.5
CMS-CPCW/H+SH [giao phiên ∧ phiên khác]	0.673 / 0.002 / 0.015 / 0.00 / 33.4
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác]	0.670 / 0.002 / 0.010 / 0.00 / 34.4
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác]	0.671 / 0.002 / 0.010 / 0.00 / 34.0

Luật	gộp
CMS-CPCW/H+SH [phiên (chuyển đạo)], α = 0.01	0.622 / 0.004 / 0.018 / 0.06 / 40.4
CMS-CPCW/H+SH+NGRr [phiên (chuyển đạo)], α = 0.01	0.619 / 0.003 / 0.017 / 0.03 / 41.3
CMS-CPCW/H+SH+NGR [phiên (chuyển đạo)], α = 0.01	0.620 / 0.003 / 0.017 / 0.03 / 40.9
CMS-CPCW/H+SH [phiên khác], α = 0.01	0.673 / 0.001 / 0.009 / 0.00 / 42.1
CMS-CPCW/H+SH+NGRr [phiên khác], α = 0.01	0.670 / 0.001 / 0.007 / 0.00 / 42.4
CMS-CPCW/H+SH+NGR [phiên khác], α = 0.01	0.671 / 0.001 / 0.007 / 0.00 / 42.3
CMS-CPCW/H+SH [online nhân quả], α = 0.01	0.675 / 0.001 / 0.004 / 0.00 / 49.8
CMS-CPCW/H+SH+NGRr [online nhân quả], α = 0.01	0.673 / 0.001 / 0.004 / 0.00 / 50.1
CMS-CPCW/H+SH+NGR [online nhân quả], α = 0.01	0.674 / 0.001 / 0.004 / 0.00 / 50.0
CMS-CPCW/H+SH [giao phiên ∧ phiên khác], α = 0.01	0.606 / 0.000 / 0.004 / 0.00 / 53.3
CMS-CPCW/H+SH+NGRr [giao phiên ∧ phiên khác], α = 0.01	0.602 / 0.000 / 0.004 / 0.00 / 53.7
CMS-CPCW/H+SH+NGR [giao phiên ∧ phiên khác], α = 0.01	0.604 / 0.000 / 0.004 / 0.00 / 53.5
CMS-CPC/H+SH+G [phiên (chuyển đạo)]	0.703 / 0.016 / 0.044 / 0.00 / 28.5
CMS-CPC/H+SH+G+NGRr [phiên (chuyển đạo)]	0.699 / 0.009 / 0.023 / 0.00 / 30.1
CMS-CPC/H+SH+G+NGR [phiên (chuyển đạo)]	0.700 / 0.009 / 0.025 / 0.00 / 29.8
CMS-CPC/H+SH+G [phiên khác]	0.719 / 0.011 / 0.035 / 0.00 / 30.4
CMS-CPC/H+SH+G+NGRr [phiên khác]	0.716 / 0.006 / 0.015 / 0.00 / 31.4
CMS-CPC/H+SH+G+NGR [phiên khác]	0.717 / 0.006 / 0.017 / 0.00 / 31.1
CMS-CPC/H+SH+G [online nhân quả]	0.711 / 0.003 / 0.020 / 0.00 / 30.9
CMS-CPC/H+SH+G+NGRr [online nhân quả]	0.708 / 0.002 / 0.009 / 0.00 / 31.1
CMS-CPC/H+SH+G+NGR [online nhân quả]	0.709 / 0.002 / 0.011 / 0.00 / 31.0
CMS-CPC/H+SH+G [giao phiên ∧ phiên khác]	0.687 / 0.009 / 0.030 / 0.00 / 31.1
CMS-CPC/H+SH+G+NGRr [giao phiên ∧ phiên khác]	0.685 / 0.006 / 0.015 / 0.00 / 31.9
CMS-CPC/H+SH+G+NGR [giao phiên ∧ phiên khác]	0.685 / 0.006 / 0.016 / 0.00 / 31.7

Bảng 10a. Bản đồ khuyết 10 % (3 seed × 6 descriptor, cổng sửa/giới hạn tốc độ)

Luật	Boreas	CMU	Nordland	4S office	4S old town	4S countryside
Xu'21 (nhãn)	0.769 / 0.076 / 0.894 / 19.9	0.705 / 0.047 / 0.160 / 5.6	0.746 / 0.027 / 0.093 / 29.3	0.728 / 0.145 / 0.958 / 12.4	0.647 / 0.072 / 0.649 / 37.5	0.278 / 0.093 / 0.976 / 60.1
Claxton'24 (nhãn)	0.785 / 0.214 / 0.939 / 33.3	0.891 / 0.263 / 0.709 / 10.4	0.552 / 0.515 / 0.763 / 65.3	0.811 / 0.208 / 0.679 / 17.2	0.754 / 0.257 / 0.699 / 85.5	0.624 / 0.262 / 0.373 / 37.2
CPC-SC+CS	0.689 / 0.000 / 0.000 / 4.4	0.758 / 0.001 / 0.006 / 3.0	0.647 / 0.000 / 0.000 / 10.7	0.719 / 0.000 / 0.000 / 9.5	0.639 / 0.000 / 0.000 / 26.0	0.305 / 0.000 / 0.000 / 50.4
CPC-SC+CS+NGRr	0.689 / 0.000 / 0.000 / 4.4	0.758 / 0.000 / 0.000 / 3.0	0.646 / 0.000 / 0.000 / 10.8	0.719 / 0.000 / 0.000 / 9.5	0.630 / 0.000 / 0.000 / 27.3	0.296 / 0.000 / 0.000 / 52.0
CMS-CPC/ H+SH	0.817 / 0.001 / 0.007 / 8.7	0.848 / 0.031 / 0.097 / 5.1	0.811 / 0.002 / 0.019 / 4.5	0.816 / 0.003 / 0.028 / 7.4	0.747 / 0.006 / 0.035 / 22.5	0.552 / 0.003 / 0.035 / 27.5
CMS-CPC/ H+SH+NGRr	0.817 / 0.000 / 0.002 / 4.6	0.846 / 0.016 / 0.079 / 3.9	0.810 / 0.001 / 0.009 / 4.4	0.816 / 0.001 / 0.014 / 7.3	0.741 / 0.004 / 0.018 / 21.2	0.542 / 0.001 / 0.012 / 29.1
CMS-CPC/ H+SH+NGR	0.817 / 0.000 / 0.002 / 8.5	0.846 / 0.016 / 0.079 / 4.3	0.810 / 0.001 / 0.009 / 4.3	0.816 / 0.001 / 0.014 / 7.3	0.743 / 0.004 / 0.018 / 21.0	0.545 / 0.002 / 0.019 / 28.3
CMS-CPC/ H+SH+G+NGRr	0.812 / 0.000 / 0.000 / 3.2	0.838 / 0.016 / 0.079 / 3.7	0.774 / 0.000 / 0.000 / 5.5	0.812 / 0.001 / 0.014 / 7.4	0.736 / 0.004 / 0.018 / 21.5	0.519 / 0.000 / 0.000 / 32.5
CMS-CPC/ H+SH+G+NGR	0.812 / 0.000 / 0.000 / 4.9	0.838 / 0.016 / 0.079 / 3.9	0.774 / 0.000 / 0.000 / 5.4	0.812 / 0.001 / 0.014 / 7.4	0.737 / 0.004 / 0.018 / 21.3	0.522 / 0.000 / 0.000 / 31.2

Luật	Boreas	CMU	Nordland	4S office	4S old town	4S countryside
CMS-CPCW/ H+SH+NGRr	0.802 / 0.000 / 0.000 / 4.4	0.841 / 0.011 / 0.067 / 3.8	0.786 / 0.000 / 0.000 / 5.1	0.810 / 0.001 / 0.007 / 7.5	0.734 / 0.004 / 0.018 / 21.8	0.496 / 0.001 / 0.008 / 35.4
CMS-CPCW/ H+SH+NGR	0.802 / 0.000 / 0.000 / 7.1	0.841 / 0.011 / 0.067 / 4.2	0.786 / 0.000 / 0.000 / 5.0	0.810 / 0.001 / 0.007 / 7.5	0.736 / 0.004 / 0.018 / 21.6	0.499 / 0.001 / 0.012 / 35.1
CMS-CPC/ H+SH+NGRr, α = 0.01	0.783 / 0.000 / 0.000 / 2.9	0.813 / 0.013 / 0.079 / 3.4	0.750 / 0.001 / 0.009 / 6.4	0.805 / 0.000 / 0.007 / 7.5	0.720 / 0.004 / 0.018 / 22.9	0.409 / 0.001 / 0.008 / 43.5
CMS-CPCW/ H+SH+NGRr, α = 0.01	0.768 / 0.000 / 0.000 / 3.4	0.797 / 0.009 / 0.067 / 3.3	0.605 / 0.000 / 0.000 / 10.5	0.761 / 0.000 / 0.000 / 8.1	0.697 / 0.004 / 0.018 / 24.3	0.355 / 0.001 / 0.008 / 57.9
Oracle	0.88 / – / – / 1.8	0.94 / – / – / 2.7	0.66 / – / – / 4.3	0.90 / – / – / 6.3	0.84 / – / – / 15.9	0.74 / – / – / 14.2

Bảng 10b. Bản đồ khuyết 50 % (3 seed × 6 descriptor, cổng sửa/giới hạn tốc độ)

Luật	Boreas	CMU	Nordland	4S office	4S old town	4S countryside
Xu'21 (nhãn)	0.715 / 0.027 / 0.071 / 32.1	0.568 / 0.080 / 0.986 / 21.9	0.754 / 0.037 / 0.094 / 121.5	0.786 / 0.381 / 1.000 / 56.0	0.559 / 0.029 / 0.047 / 98.8	0.272 / 0.221 / 0.968 / 156.2
Claxton'24 (nhãn)	0.716 / 0.065 / 0.201 / 33.5	0.796 / 0.071 / 0.228 / 16.4	0.509 / 0.132 / 0.207 / 89.5	0.768 / 0.086 / 0.285 / 25.0	0.714 / 0.087 / 0.261 / 133.2	0.571 / 0.099 / 0.198 / 47.5
CPC-SC+CS	0.600 / 0.000 / 0.007 / 5.8	0.609 / 0.012 / 0.090 / 6.9	0.669 / 0.000 / 0.001 / 13.7	0.697 / 0.000 / 0.000 / 15.5	0.572 / 0.000 / 0.000 / 41.0	0.250 / 0.001 / 0.011 / 59.5
CPC-SC+CS+NGRr	0.600 / 0.000 / 0.004 / 5.4	0.609 / 0.009 / 0.071 / 5.5	0.662 / 0.000 / 0.001 / 13.9	0.694 / 0.000 / 0.000 / 15.6	0.555 / 0.000 / 0.000 / 42.6	0.232 / 0.001 / 0.006 / 62.1
CMS-CPC/ H+SH	0.736 / 0.005 / 0.016 / 11.6	0.759 / 0.053 / 0.187 / 26.7	0.799 / 0.011 / 0.049 / 25.0	0.791 / 0.003 / 0.016 / 14.9	0.717 / 0.011 / 0.024 / 48.3	0.492 / 0.010 / 0.026 / 45.8
CMS-CPC/ H+SH+NGRr	0.735 / 0.002 / 0.010 / 6.0	0.758 / 0.028 / 0.126 / 10.4	0.786 / 0.002 / 0.012 / 11.5	0.788 / 0.001 / 0.008 / 13.6	0.704 / 0.007 / 0.022 / 36.4	0.475 / 0.006 / 0.017 / 40.3
CMS-CPC/ H+SH+NGR	0.735 / 0.002 / 0.010 / 8.3	0.758 / 0.030 / 0.130 / 17.3	0.794 / 0.003 / 0.018 / 14.6	0.788 / 0.001 / 0.008 / 14.0	0.708 / 0.007 / 0.022 / 37.0	0.480 / 0.006 / 0.022 / 44.4
CMS-CPC/ H+SH+G+NGRr	0.730 / 0.002 / 0.009 / 5.2	0.743 / 0.027 / 0.125 / 9.9	0.766 / 0.000 / 0.003 / 9.6	0.783 / 0.001 / 0.007 / 13.6	0.696 / 0.003 / 0.009 / 36.9	0.457 / 0.004 / 0.015 / 41.9
CMS-CPC/ H+SH+G+NGR	0.730 / 0.002 / 0.010 / 6.8	0.742 / 0.028 / 0.128 / 15.9	0.771 / 0.000 / 0.004 / 8.9	0.783 / 0.001 / 0.007 / 13.8	0.700 / 0.003 / 0.009 / 37.4	0.462 / 0.005 / 0.019 / 42.1
CMS-CPCW/ H+SH+NGRr	0.721 / 0.001 / 0.009 / 5.7	0.745 / 0.027 / 0.125 / 9.9	0.770 / 0.000 / 0.000 / 9.5	0.782 / 0.001 / 0.004 / 13.8	0.694 / 0.006 / 0.014 / 36.6	0.443 / 0.005 / 0.013 / 43.0
CMS-CPCW/ H+SH+NGR	0.721 / 0.001 / 0.009 / 7.5	0.745 / 0.029 / 0.127 / 16.0	0.773 / 0.000 / 0.000 / 9.0	0.782 / 0.001 / 0.004 / 14.0	0.698 / 0.006 / 0.014 / 36.7	0.449 / 0.005 / 0.015 / 42.6
CMS-CPC/ H+SH+NGRr, α = 0.01	0.712 / 0.001 / 0.003 / 5.3	0.707 / 0.018 / 0.109 / 7.1	0.736 / 0.001 / 0.007 / 10.5	0.712 / 0.001 / 0.008 / 14.7	0.680 / 0.007 / 0.022 / 37.3	0.390 / 0.003 / 0.011 / 48.1
CMS-CPCW/ H+SH+NGRr, α = 0.01	0.695 / 0.000 / 0.001 / 5.0	0.688 / 0.017 / 0.110 / 6.9	0.643 / 0.000 / 0.000 / 13.7	0.631 / 0.000 / 0.004 / 15.9	0.648 / 0.005 / 0.012 / 39.8	0.335 / 0.002 / 0.011 / 59.2
Oracle	0.89 / – / – / 2.5	0.90 / – / – / 3.3	0.69 / – / – / 8.0	0.91 / – / – / 12.0	0.85 / – / – / 31.8	0.73 / – / – / 23.6

Bảng 9a. LODO, α = 0.05

datasets ['4s:countryside', '4s:office_loop', '4s:old_town', 'boreas', 'cmu', 'nclt', 'nord'] candidates ['CMS-CPC/H+SH', 'CMS-CPC/H+SH+G', 'CMS-CPC/H+SH+G+NGR', 'CMS-CPC/H+SH+G+NGRr', 'CMS-CPC/H+SH+G[both]', 'CMS-CPC/H+SH+G[both]+NGR', 'CMS-CPC/H+SH+G[both]+NGRr', 'CMS-CPC/H+SH+G[loco]', 'CMS-CPC/H+SH+G[loco]+NGR', 'CMS-CPC/H+SH+G[loco]+NGRr', 'CMS-CPC/H+SH+G[online]', 'CMS-CPC/H+SH+G[online]+NGR', 'CMS-CPC/H+SH+G[online]+NGRr',

'CMS-CPC/H+SH+NGR', 'CMS-CPC/H+SH+NGRr', 'CMS-CPCW/H+SH', 'CMS-CPCW/H+SH+NGR', 'CMS-CPCW/H+SH+NGRr', 'CMS-CPCW/H+SH[both]', 'CMS-CPCW/H+SH[both]+NGR', 'CMS-CPCW/H+SH[both]+NGRr', 'CMS-CPCW/H+SH[loco]', 'CMS-CPCW/H+SH[loco]+NGR', 'CMS-CPCW/H+SH[loco]+NGRr', 'CMS-CPCW/H+SH[online]', 'CMS-CPCW/H+SH[online]+NGR', 'CMS-CPCW/H+SH[online]+NGRr', 'CPC-SC+CS', 'CPC-SC+CS+NGRr', 'CPC-SC-T0.5+SH', 'CPC-SC-T0.5+SH+NGR', 'CPC-SC-T0.5+SH+NGRr']

held-out	chosen on the others	held-out err	recall	segFAR	best label-free on held-out	regret (m)
4s:countryside	CMS-CPC/H+SH+NGRr	34.3	0.491	0.001	CMS-CPC/H+SH+NGRr (31.8)	2.5
4s:office_loop	CMS-CPC/H+SH+NGRr	9.7	0.800	0.000	CMS-CPC/H+SH+NGRr (9.6)	0.1
4s:old_town	CMS-CPC/H+SH+NGRr	26.1	0.703	0.006	CMS-CPC/H+SH+NGR (25.8)	0.3
boreas	CMS-CPC/H+SH+NGRr	4.5	0.775	0.003	CMS-CPC/H+SH[both]+NGRr (4.3)	0.2
cmu	CMS-CPC/H+SH+NGRr	4.0	0.822	0.009	CPC-SC+CS+NGRr (3.4)	0.6
nclt	CMS-CPC/H+SH+NGRr	4.3	0.612	0.002	CMS-CPCW/H+SH[online]+NGRr (3.7)	0.6
nord	CMS-CPC/H+SH+NGRr	5.6	0.765	0.000	CMS-CPC/H+SH+NGRr (5.5)	0.1

Bảng 9b. LODO, α = 0.01

datasets ['4s:countryside', '4s:office_loop', '4s:old_town', 'boreas', 'cmu', 'nclt', 'nord'] candidates ['CMS-CPC/H+SH(a=0.01)+NGR', 'CMS-CPC/H+SH(a=0.01)+NGRr', 'CMS-CPC/H+SH+G(a=0.01)', 'CMS-CPC/H+SH+G(a=0.01)+NGR', 'CMS-CPC/H+SH+G(a=0.01)+NGRr', 'CMS-CPC/H+SH+Gboth', 'CMS-CPC/H+SH+Gboth+NGR', 'CMS-CPC/H+SH+Gboth+NGRr', 'CMS-CPC/H+SH+Gloco', 'CMS-CPC/H+SH+Gloco+NGR', 'CMS-CPC/H+SH+Gloco+NGRr', 'CMS-CPC/H+SH+Gonline', 'CMS-CPC/H+SH+Gonline+NGR', 'CMS-CPC/H+SH+Gonline+NGRr', 'CMS-CPCW/H+SH(a=0.01)', 'CMS-CPCW/H+SH(a=0.01)+NGR', 'CMS-CPCW/H+SH(a=0.01)+NGRr', 'CMS-CPCW/H+SHboth', 'CMS-CPCW/H+SHboth+NGR', 'CMS-CPCW/H+SHboth+NGRr', 'CMS-CPCW/H+SHloco', 'CMS-CPCW/H+SHloco+NGR', 'CMS-CPCW/H+SHloco+NGRr', 'CMS-CPCW/H+SHonline', 'CMS-CPCW/H+SHonline+NGR', 'CMS-CPCW/H+SHonline+NGRr']

held-out	chosen on the others	held-out err	recall	segFAR	best label-free on held-out	regret (m)
4s:countryside	CMS-CPC/H+SH+G(a=0.01)+NGRr	46.2	0.384	0.000	CMS-CPC/H+SH+G(a=0.01) (43.8)	2.4
4s:office_loop	CMS-CPC/H+SH+G(a=0.01)+NGRr	10.8	0.726	0.000	CMS-CPC/H+SHloco (10.3)	0.4
4s:old_town	CMS-CPC/H+SH+G(a=0.01)+NGRr	27.1	0.685	0.004	CMS-CPC/H+SH+G(a=0.01)+NGR (26.7)	0.4
boreas	CMS-CPC/H+SH+G(a=0.01)+NGR	4.1	0.748	0.001	CMS-CPC/H+SH+G(a=0.01)+NGRr (3.9)	0.2
cmu	CMS-CPC/H+SH+G(a=0.01)+NGRr	3.6	0.786	0.007	CMS-CPC/H+SHonline+NGRr (3.5)	0.0
nclt	CMS-CPC/H+SH+G(a=0.01)+NGRr	3.9	0.581	0.001	CMS-CPC/H+SH+G(a=0.01)+NGRr (3.9)	0.0
nord	CMS-CPC/H+SH+G(a=0.01)+NGRr	7.3	0.694	0.000	CMS-CPC/H+SH+G(a=0.01)+NGR (7.2)	0.1

5. Đọc kết quả

Hiệu chuẩn nhân quả (Bảng 1–4). Bản chuyển đạo (`cond`) vẫn có recall cao nhất trên benchmark chính (dữ liệu của phiên giúp khớp điều kiện). Trong các bản nhân quả, `[both]` (giao phiên ∧ phiên khác) tốt nhất và ổn định nhất: hơn CPC-SC-T½+SH 7/7 cả hai chỉ số. `[online]` (phiên khác + phần đã qua của phiên) mất recall ở Nordland (0.695 so với 0.713): ở đầu mỗi phiên tập hiệu chuẩn chưa có gì của điều kiện hiện tại, và Nordland có điều kiện mùa khác hẳn nhau. Giá của tính nhân quả so với bản chuyển đạo: 0.01–0.05 recall, sai số gần như không đổi (nhờ cổng).

Hai biến thể cổng (Bảng 4). Mở cả chuỗi (+NGR) để lọt các chuỗi báo sai đồng nhất: Boreas +2.0 đến +2.9 m, CMU +0.5 m (CI > 0); chỉ có lợi ở 4S countryside (−0.5 đến −1.0 m, VIO trôi). Mở giới hạn tốc độ (+NGRr) chặn tác hại của chuỗi sai đồng nhất ở K·ν mỗi R frame. LODO chọn +NGRr 7/7.

Độ bền bộ lọc (Bảng 5–6). Kết luận chính không phụ thuộc bộ lọc: dưới F1, F2, F3 phương pháp đề xuất hơn luật cũ 7/7 cả recall lẫn sai số. Cổng giảm sai số 7/7 dưới F1/F2 nhưng **không** giúp dưới F3 (2/7): khi bộ lọc khởi tạo sai, innovation của fix đúng lớn và cổng (giả định bộ lọc đã định vị) giữ lại chúng — sàn $3\sqrt{P^{-}}$ chỉ mở cổng trong vài bước đầu khi P còn lớn. Đây là giới hạn đúng như thiết kế của cổng, không phải của CMS (bản không cổng vẫn hơn luật cũ dưới F3).

Off-map tự nhiên (Bảng 7–8). Đây là phép thử quan trọng nhất về tính hợp lệ: phiên của cảnh B hoàn toàn off-map nên nhóm `cond` có $\pi = 0$ trong phiên đó. Thực tế ở $\alpha = 0.05$ mọi biến thể vẫn hợp lệ (segFAR tệ nhất 0.044 với CMS-CPC/H+SH+G không cổng, 0.013 với cổng +NGRr): nhiễm của frame off-map yếu vì xoá một hành lang chỉ bỏ một phần nhỏ các khớp giả. Ở $\alpha = 0.01$ bản `cond` vượt α ở 2/36 run không cổng, 1/36 với +NGRr (đúng dự báo của Mệnh đề 8), các bản nhân quả không run nào. Recall cảnh A: CMS 0.67–0.72 so với 0.59 của luật cũ (6/6 cặp). Baseline có nhãn (được xem một bản ghi có nhãn của cảnh B): Claxton'24 recall cao hơn ở hai cặp countryside nhưng vượt α ở 14 % run; Xu'21 28 %; TAO'26 11 %. ⇒ **Khuyến nghị triển khai:** `[online]` hoặc `[both]` (không bao giờ vượt α ở đây), mất ≈ 0.01–0.05 recall so với bản chuyển đạo.

HMM liên hợp (Entry 31, §2 lý thuyết §4d). Kết quả thiết kế (3 run): xem `RESEARCH_LOG.md` Entry 31. Đánh giá 114 run (seed 0–2, 7 dataset × 6 descriptor) đang chạy trên 20 phiên L4; bảng E31 sẽ được thêm vào báo cáo này khi xong.

6. Hạn chế và việc tiếp theo

- Benchmark tự nhiên chỉ dùng 3 cảnh 4Seasons (6 cặp); cảnh B khác thành phố nhưng cùng loại ảnh (camera, mùa). Off-map "gần" (đi lạc sang phố bên cạnh bản đồ) vẫn chỉ có ở lỗ tổng hợp và NCLT.
- Sai số điều hướng tuyệt đối vẫn phụ thuộc bộ lọc đánh giá; F1–F3 cho thấy **so sánh** giữa các luật bền, không phải mức tuyệt đối.
- Cổng +NGRr giả định bộ lọc đã định vị (F3); chọn giữa hai biến thể mở khoá dựa trên LODO, không có định lý về tác hại.
- `[online]` cần các phiên trước có phần on-map; với robot ở phiên đầu tiên (chưa có phiên nào khác) chỉ còn bản chuyển đạo hoặc null bản đồ.
- Việc tiếp theo: dữ liệu MarsLab (khi có máy RTX ghi rosbag) cho off-map tự nhiên kiểu "đi ra ngoài vùng bản đồ" và vòng kín điều khiển.